

PubChem Structure Validation — Analysis Report

ModelSEEDDatabase · Validate_PubChem_Structures.py · 2026-06-01

Result: 68 existing compound structures corrected, 40 new compounds resolved, 3 removed.
Phase 5 applied 111 structural corrections (70 STEREO_DIFF, 20 PROTONATION_DIFF, 21 pKa);
109 produced a visible before/after change (the remainder land on newly-added compounds).
415 further SMILES were re-canonicalized with no chemical change (identity preserved).

Change-type breakdown (68 corrected)

change type	count
Stereochemistry gained	48
Stereo/representation change	13
Protonation + stereo gain	6
Protonation change	1

Reaction-usage context

set	% orphan	median	mean	max
Corrected (68)	44%	1	2.1	42
All structured cpds	38%	1	6.3	28,815
Newly added (40)	10%	3	6.8	65

Corrected compounds are peripheral secondary-metabolites; heavily-used core metabolites were already correct and untouched.

Pipeline (Phases 0-6)

Phase 0 Self-consistency

Parse SMILES; compute missing InChI/InChIKey; verify InChIKey and SMILES/InChI connectivity; recompute formula & charge from InChI. Skip R-group/unparseable.

Phase 1 External-ID lookup

ChEBI/KEGG aliases -> PubChem CID via xref. xref conflicts scored against stored InChIKey.

Phase 2 Name lookup

Batch name -> CID; disambiguate multi-hit by InChIKey blocks then Tanimoto.

Phase 3 Recovery

Direct InChIKey then InChI lookup for still-unresolved/mismatched.

Phase 4 Classification

MATCH / PROTONATION_DIFF / STEREO_DIFF / MISMATCH / NOT_FOUND / XREF_CONFLICT / AMBIGUOUS; sub-classify mismatches (tautomer/salt/metal/wrong-mapping/isomer).

Phase 5 Corrections

STEREO_DIFF (PubChem >= stored stereo, no inversion incl. /m enantiomer guard, consistent SMILES/InChIKey); pKa protonation at pH7 (borderline 6-8 skipped, strong-acid & metal-complex guards). Rejections logged.

Phase 6 Apply & write

Write corrections to All file; recompute formula/charge; canonicalize all SMILES; rebuild Unique via dev-branch picker (InChI>SMILE, multi-DB agreement, source priority); post-correction consistency check; render before/after 2D diagrams.

Items for manual review

Manual review: cpd35694 (R)-2-(phosphomethyl)malate — PubChem (R) matches the name, stored (S) likely wrong (kept stored conservatively). cpd26222 DDF — inversion, no name descriptor. cpd23840 ABTS — radical InChI re-parse warning (pre-existing).

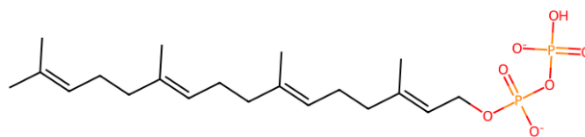
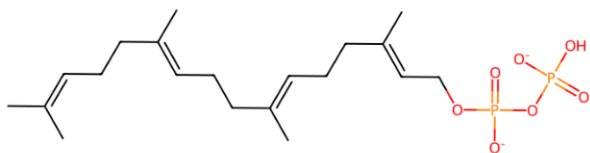
Corrected compounds (visible change) — before / after (1-3 of 58)

cpd39967 3,7,11,15-tetramethylhexadeca-2,6,10,14-tetraen-1-yl diphosphate (9 rxns)

Correction: Stereochemistry added (stereocenters 0->3)

0INNEUNVOZHBOX-UHFFFAOYSA-L -> 0INNEUNVOZHBOX-QIRCYJPOSA-L
BEFORE

AFTER

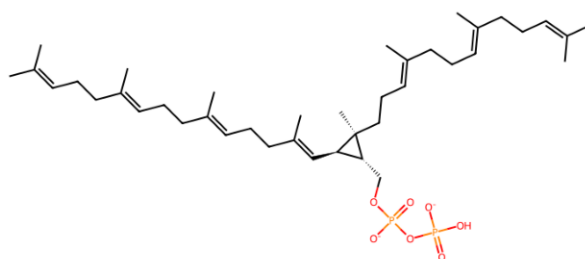
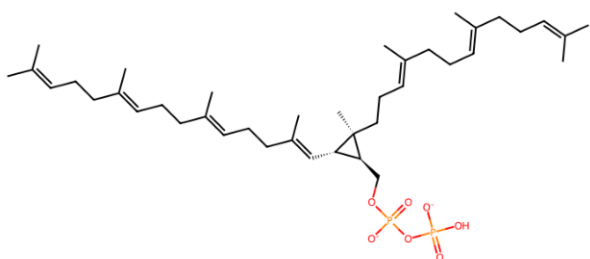


cpd02168 (1R,2R,3R)-prephytoene diphosphate (8 rxns)

Correction: Stereochemistry / representation change

RVCNKTPCHZNAAO-IMSLGMFESA-L -> RVCNKTPCHZNAAO-UZDKSQMHSA-L
BEFORE

AFTER

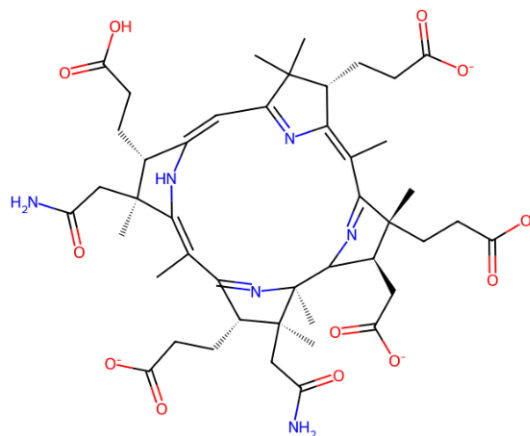
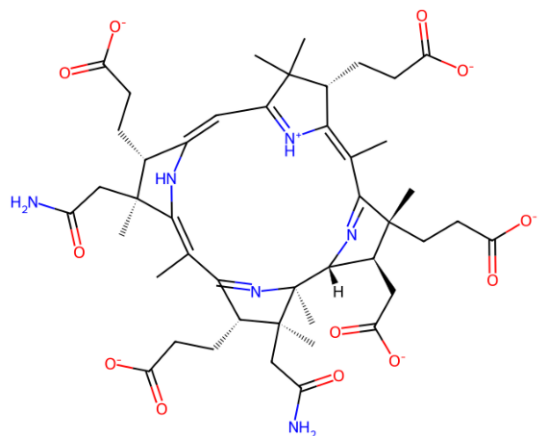


cpd03913 Hydrogenobyrrinate a,c diamide (6 rxns)

Correction: Stereochemistry added (stereocenters 10->11)

ZGGWTIPDU0THRA-FVLIVPSMSA-J -> ZGGWTIPDU0THRA-QQHNNQLSA-J
BEFORE

AFTER

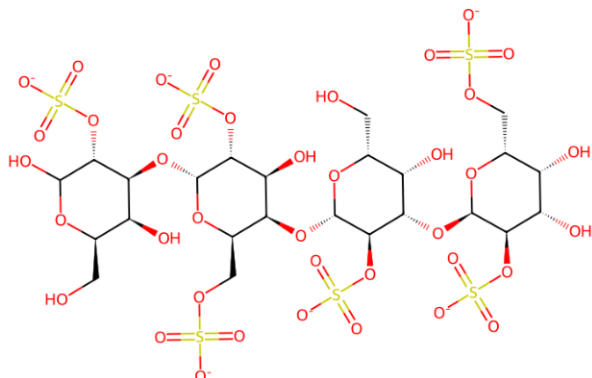


Corrected compounds (visible change) — before / after (4-6 of 58)

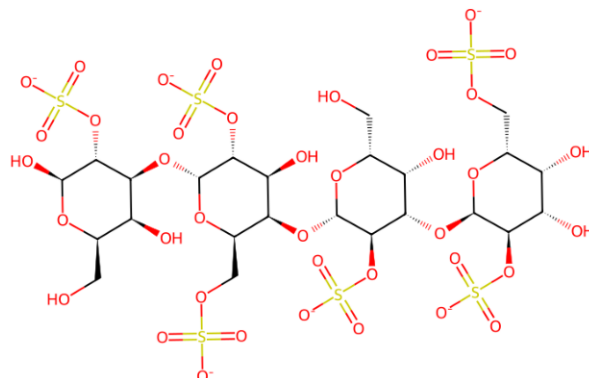
cpd23886 neo-lambda-carratetraose (5 rxns)

Correction: Stereochemistry added (stereocenters 19->20)

SYQAUHQZRLJKJT-GTBSAFNRS-A-H -> SYQAUHQZRLJKJT-JY0YXRFGS-A-H
BEFORE



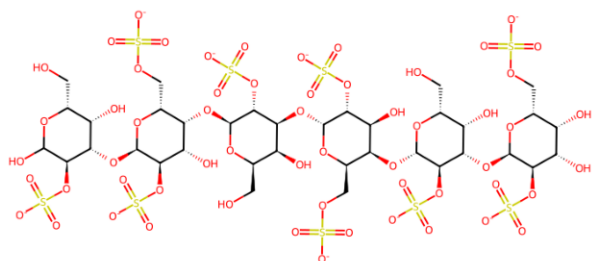
AFTER



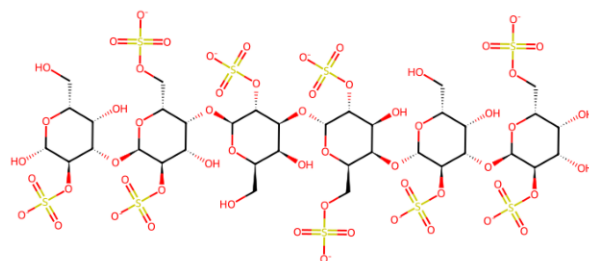
cpd23887 neo-lambda-carrahexaose (5 rxns)

Correction: Stereochemistry added (stereocenters 29->30)

XCAVJOBVNSMNME-WRUHAMN-A-E -> XCAVJOBVNSMNME-HQV0YLOSA-E
BEFORE



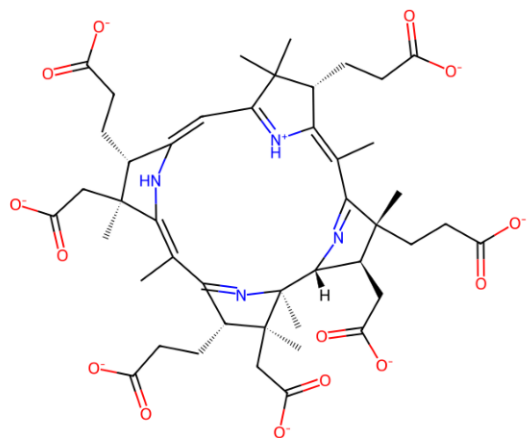
AFTER



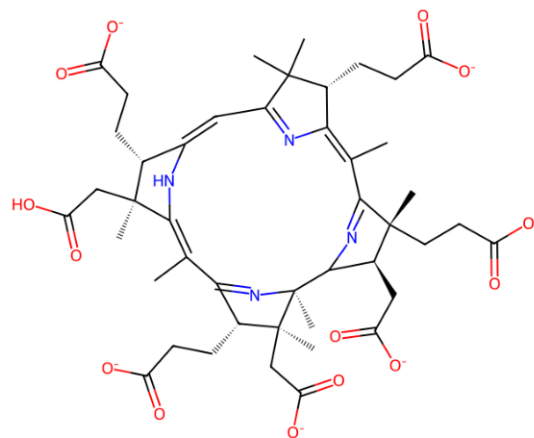
cpd03832 Hydrogenobyrrinate (4 rxns)

Correction: Stereochemistry added (stereocenters 10->11)

FJDBIDBCPIOVPG-FVLIVPSM-A-H -> FJDBIDBCPIOVPG-QQHNNQLSA-H
BEFORE



AFTER

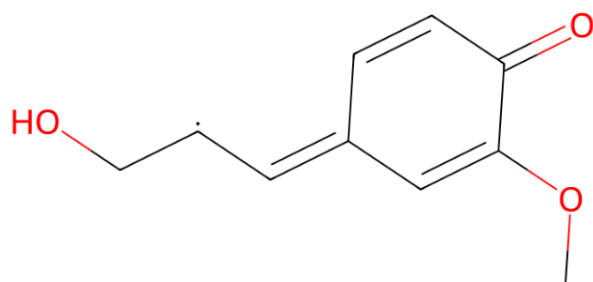


Corrected compounds (visible change) — before / after (7-9 of 58)

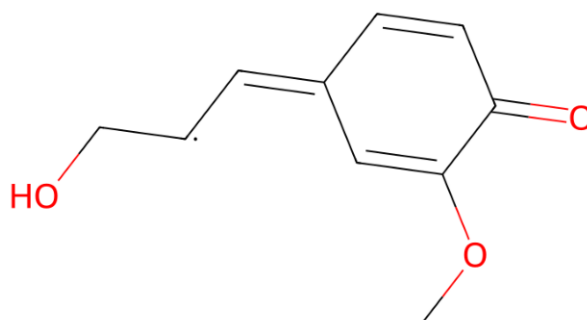
cpd35693 coniferyl alcohol radical (4 rxns)

Correction: Stereochemistry / representation change

ORAJWSYKRGVTD - FPYGCLRLSA - N -> ORAJWSYKRGVTD - UHFFFAOYSA - N
BEFORE



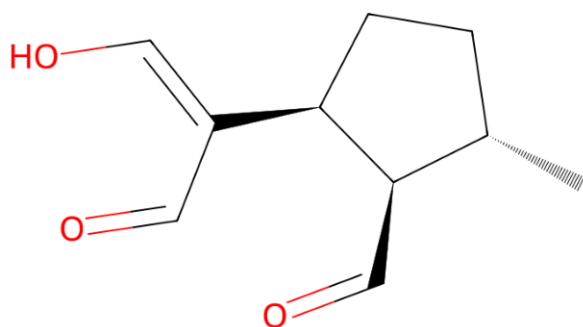
AFTER



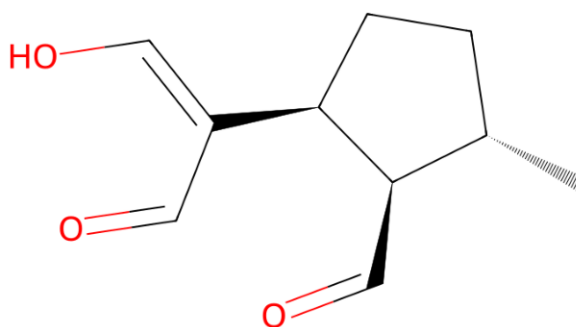
cpd03617 (1S,2R,3S)-2-Formyl-alpha-(hydroxymethylene)-3-methylcyclo (3 rxns)

Correction: Stereochemistry added (stereocenters 3->4)

PFGBAVLSGZLAAY - FXBDTBDDSA - N -> PFGBAVLSGZLAAY - JYPFKNLRSA - N
BEFORE



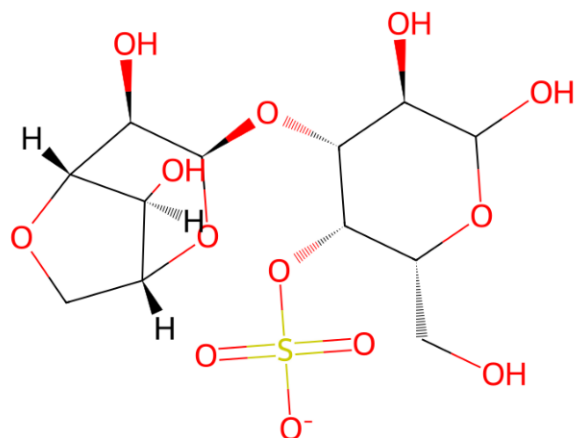
AFTER



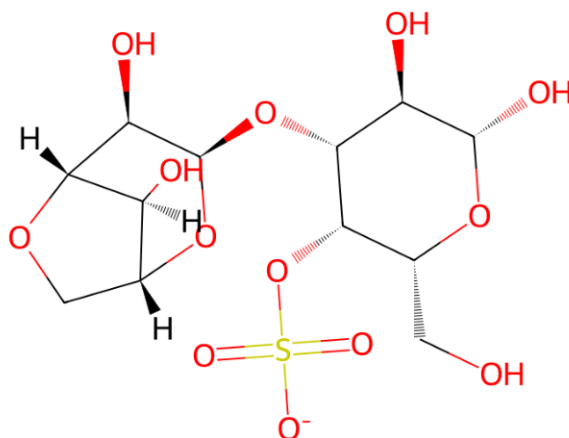
cpd23880 3,6-anhydro-alpha-D-galactopyranosyl-(1->3)-4-O-sulfo-D-ga (3 rxns)

Correction: Stereochemistry added (stereocenters 9->10)

OGFGYTRGDDWUKC - PBFIDDPKSA - M -> OGFGYTRGDDWUKC - YBFCQ00YSA - M
BEFORE



AFTER



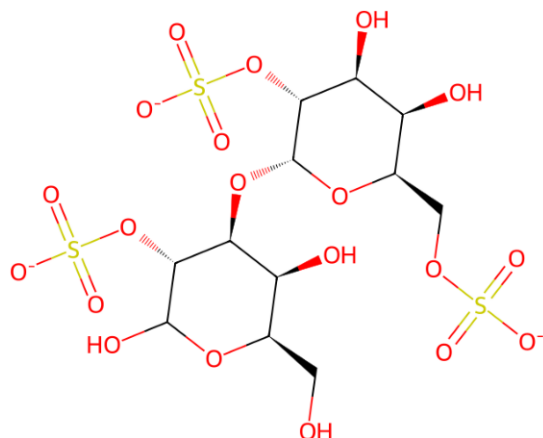
Corrected compounds (visible change) — before / after (10-12 of 58)

cpd23888 neo-lambda-carrabiose (3 rxns)

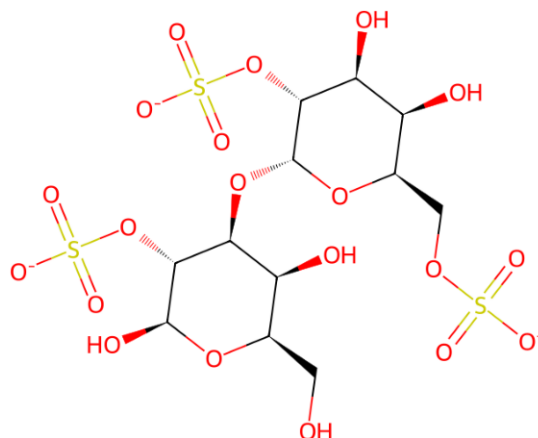
Correction: Stereochemistry added (stereocenters 9->10)

RHDWPJKPWCDPBP-JZSVMVJISA-K -> RHDWPJKPWCDPBP-WSWMMNSNSA-K

BEFORE



AFTER

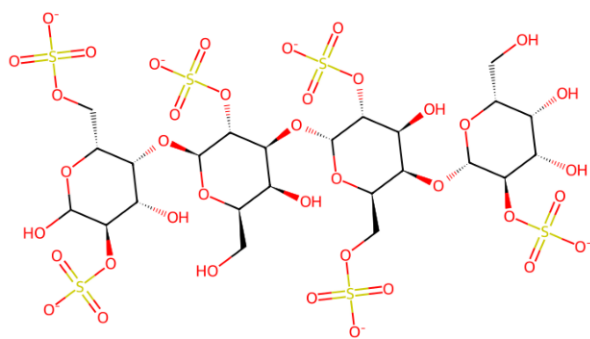


cpd25687 alpha-D-Galp2,6S2-(1->3)-beta-D-Galp2S-(1->4)-alpha-D-Galp (3 rxns)

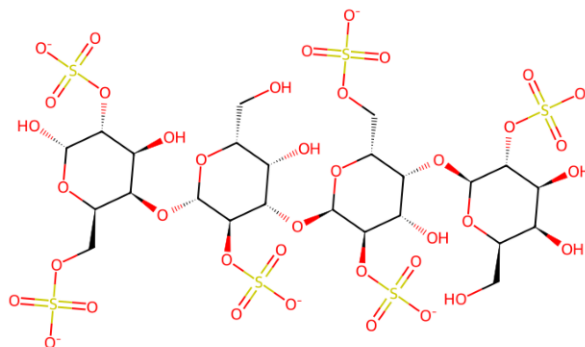
Correction: Stereochemistry added (stereocenters 19->20)

CFLYIPGVTFQOCI-SDNHUNQPSA-H -> CFLYIPGVTFQOCI-SQESNNFJSA-H

BEFORE



AFTER

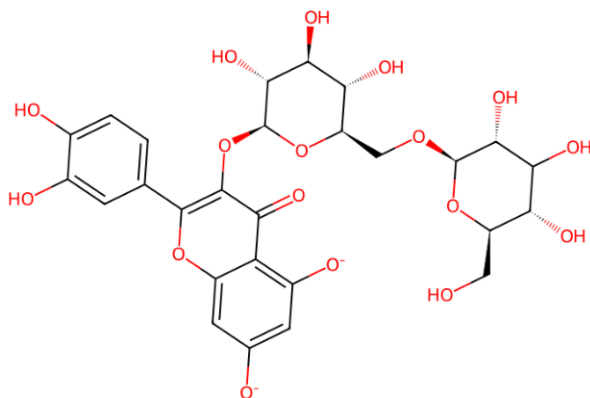


cpd31674 quercetin-3-gentiobioside (3 rxns)

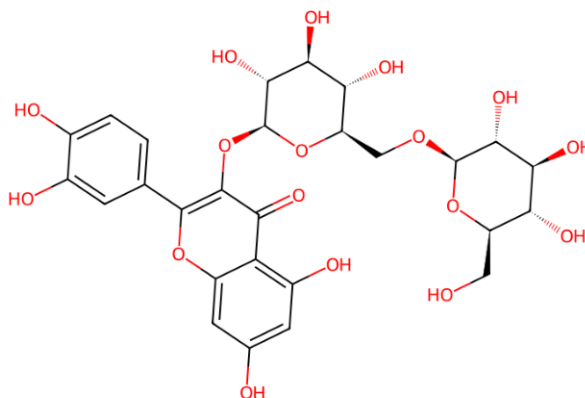
Correction: Protonation + stereochemistry (charge -2 -> 0; stereocenters 9->10)

FDRQPMVGJQVTL-BVYPKUFKSA-L -> FDRQPMVGJQVTL-DEFKTLLOSSA-N

BEFORE



AFTER

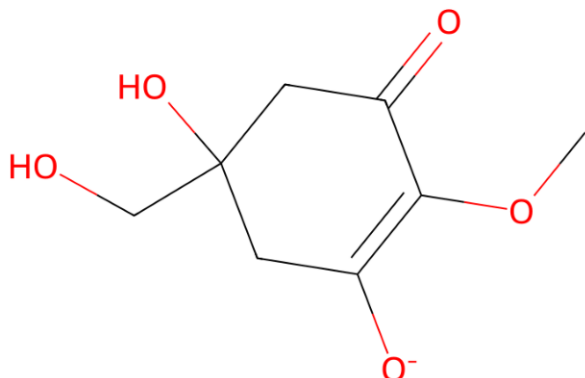


Corrected compounds (visible change) — before / after (13-15 of 58)

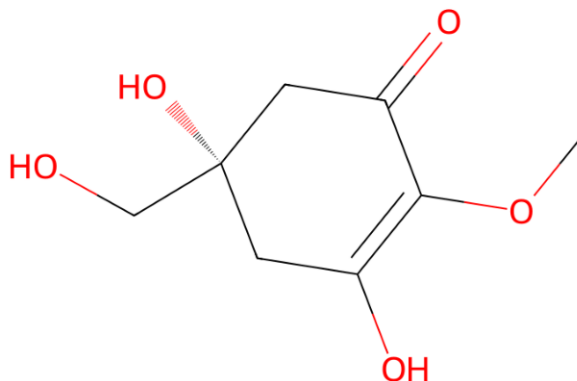
cpd32789 (S)-4-deoxygadusol (3 rxns)

Correction: Protonation + stereochemistry (charge -1 -> 0; stereocenters 0->1)

ZONRIYAALKITGT-UHFFFAOYSA-M -> ZONRIYAALKITGT-QMMMGPBSA-N
BEFORE



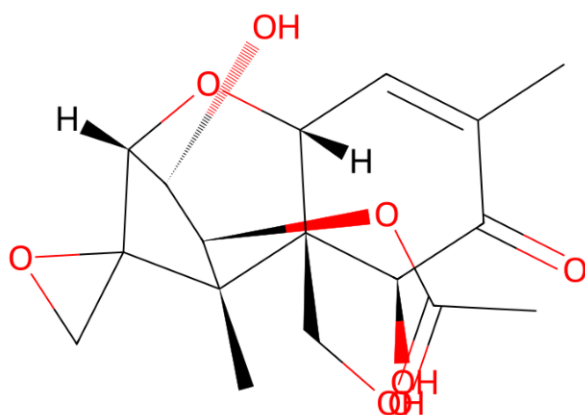
AFTER



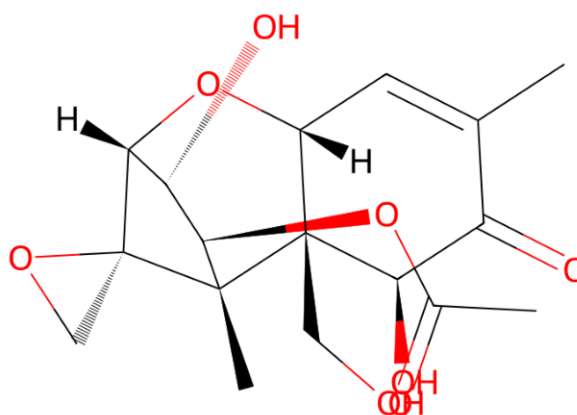
cpd35178 4-acetylinalenol (3 rxns)

Correction: Stereochemistry added (stereocenters 7->8)

XGCUCFKWVIWWN-IZEYKREJSA-N -> XGCUCFKWVIWWN-CAYGJDLQSA-N
BEFORE



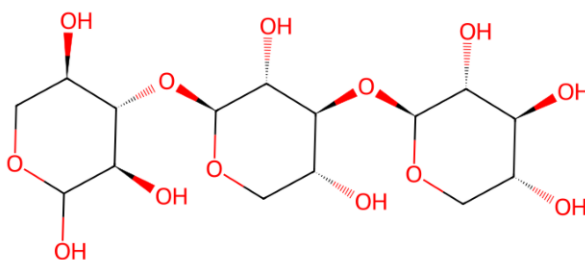
AFTER



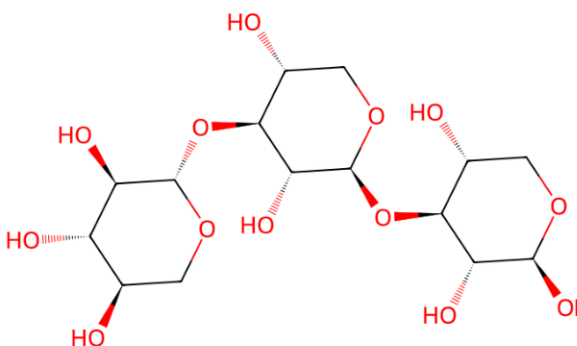
cpd23818 (1,3)-beta-xylotriose (2 rxns)

Correction: Stereochemistry added (stereocenters 11->12)

WKSXOLNEYANUQB-RTAFPKLOSA-N -> WKSXOLNEYANUQB-KWGKWNPMISA-N
BEFORE



AFTER



Corrected compounds (visible change) — before / after (16-18 of 58)

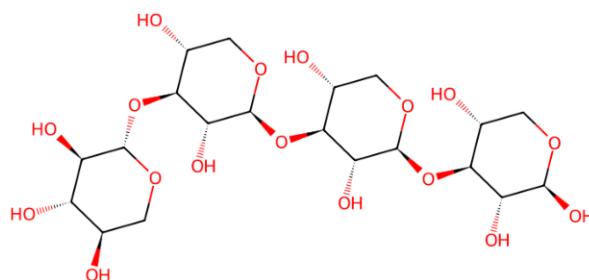
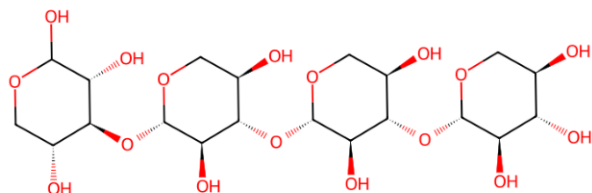
cpd23819 (1,3)-beta-xylotetraose (2 rxns)

Correction: Stereochemistry added (stereocenters 15->16)

YIMKISJWNJZSQI-GKDCLMHXSA-N -> YIMKISJWNJZSQI-JT0FAPFQSA-N

BEFORE

AFTER



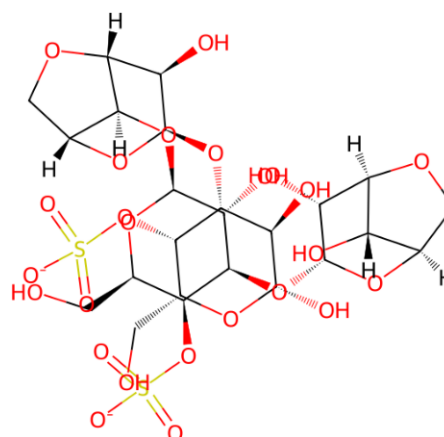
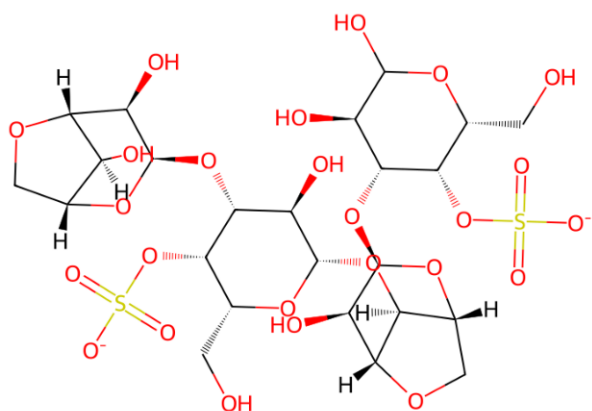
cpd23881 neocarratetraose 4-O-disulfate (2 rxns)

Correction: Stereochemistry added (stereocenters 19->20)

PPWBJIYNGRTPNY-SAEYTFRYSA-L -> PPWBJIYNGRTPNY-0KPLKWSVSA-L

BEFORE

AFTER



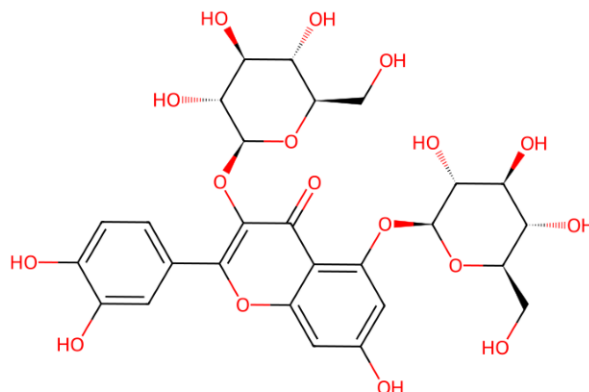
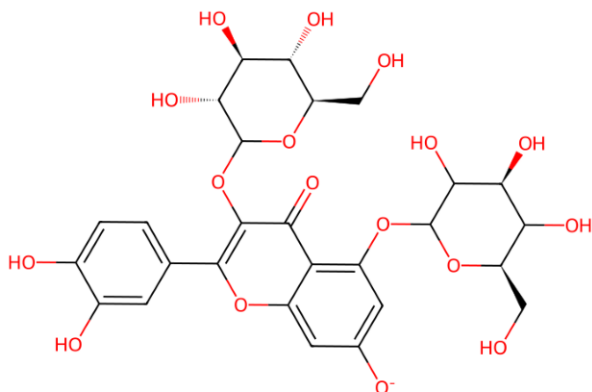
cpd25005 quercetin 3,5-O-diglucoside (2 rxns)

Correction: Protonation + stereochemistry (charge -1 -> 0; stereocenters 6->10)

YOXWSUCVDVXAMX-LOZUMUMPSA-M -> YOXWSUCVDVXAMX-DEFKTLLOSSA-N

BEFORE

AFTER



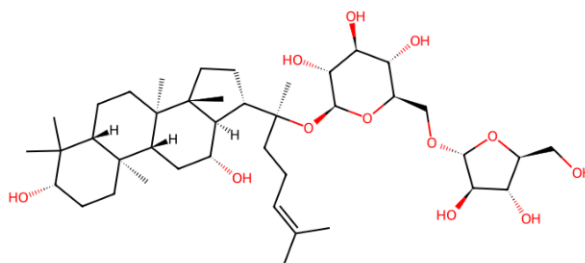
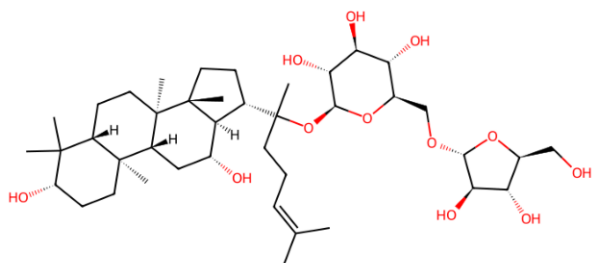
Corrected compounds (visible change) — before / after (19-21 of 58)

cpd32048 20-[alpha-L-arabinofuranosyl-(1->6)-beta-D-glucopyranosylo (2 rxns)

Correction: Stereochemistry added (stereocenters 18->19)

CJFGBCWGOQRURQ - NMUMGEHWSA - N -> CJFGBCWGOQRURQ - JFJIKBJRSA - N
BEFORE

AFTER

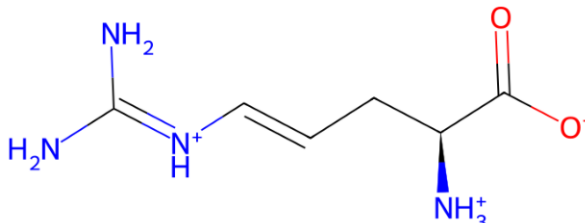
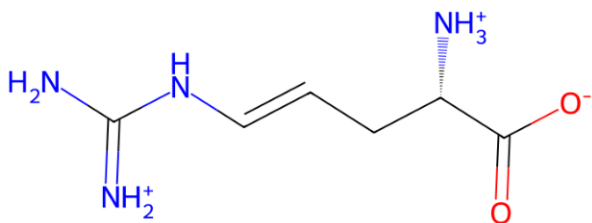


cpd32552 4,5-dehydro-L-arginine (2 rxns)

Correction: Stereochemistry added (stereocenters 1->2)

FVSQXMGZDMFDEJ - BYPYZUCNSA - 0 -> FVSQXMGZDMFDEJ - REJIZHKJSA - 0
BEFORE

AFTER

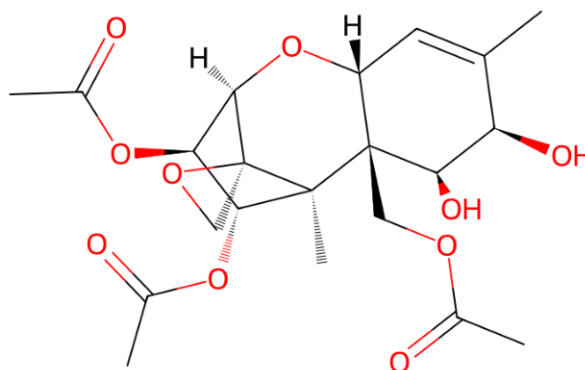
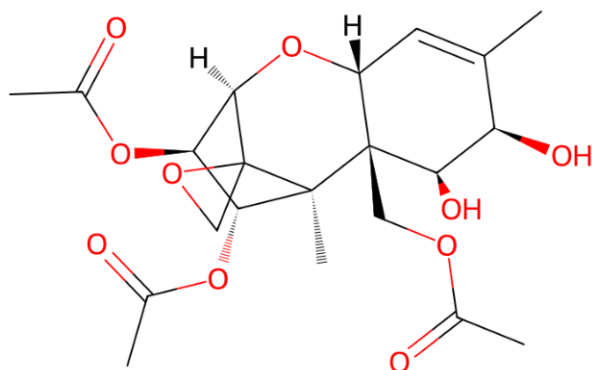


cpd33745 7,8-dihydroxy-3,4,15-triacetoxyscirpenol (2 rxns)

Correction: Stereochemistry added (stereocenters 8->9)

URJKZBKWIWMEQI - LYTWTWTFSA - N -> URJKZBKWIWMEQI - IGZVRCOWSA - N
BEFORE

AFTER

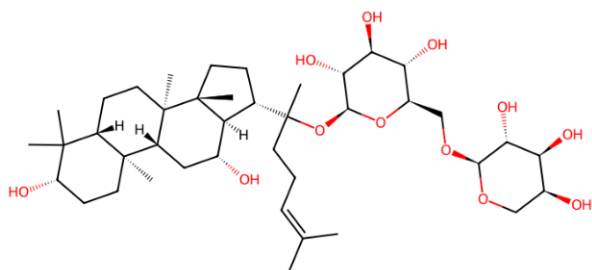


Corrected compounds (visible change) — before / after (22-24 of 58)

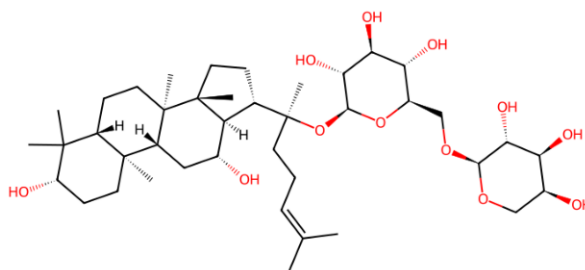
cpd34103 (20S)-ginsenoside C-Y (2 rxns)

Correction: Stereochemistry added (stereocenters 18->19)

YNBYFOIDLBTOMW-MOBYHOAZSA-N -> YNBYFOIDLBTOMW-QHNUHGIDSA-N
BEFORE



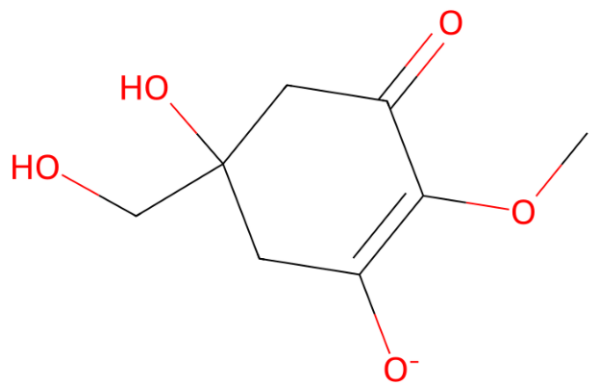
AFTER



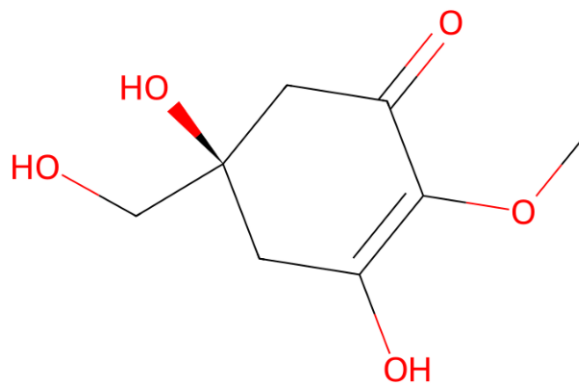
cpd34182 (R)-4-deoxygadusol (2 rxns)

Correction: Protonation + stereochemistry (charge -1 -> 0; stereocenters 0->1)

ZONRIYAALKITGT-UHFFFAOYSA-M -> ZONRIYAALKITGT-MRVPVSSYSA-N
BEFORE



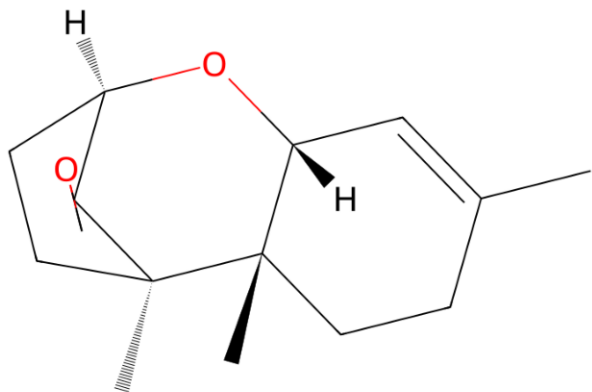
AFTER



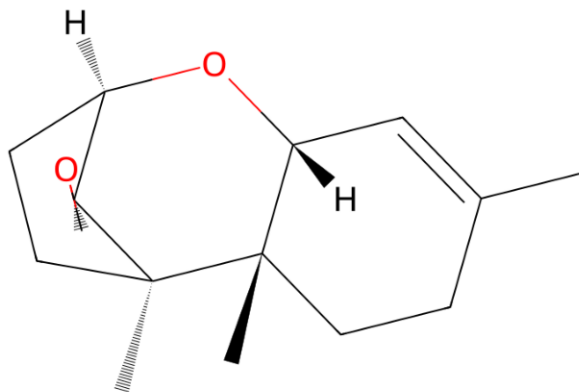
cpd34487 12,13-epoxytrichothec-9-ene (2 rxns)

Correction: Stereochemistry added (stereocenters 4->5)

LZAJKCZTKKKZNT-HHGGZCDHSA-N -> LZAJKCZTKKKZNT-QMIVQANSA-N
BEFORE



AFTER



Corrected compounds (visible change) — before / after (25-27 of 58)

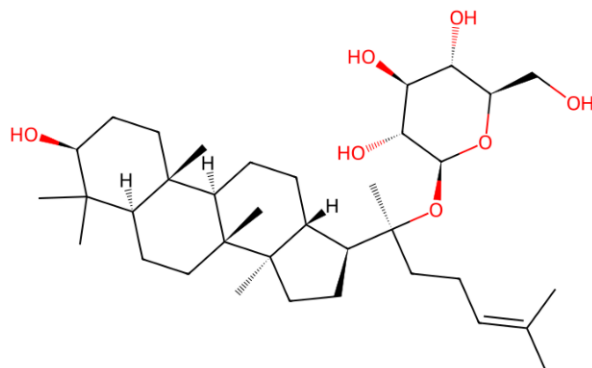
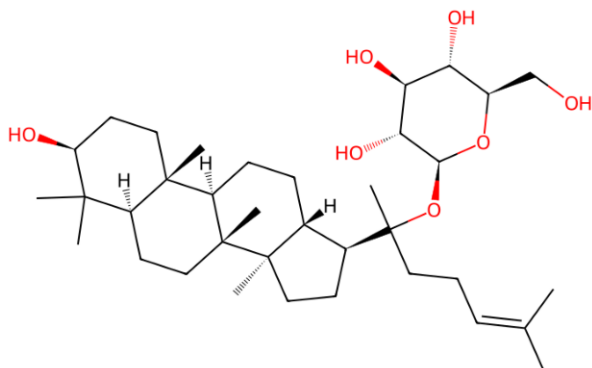
cpd38198 (20S)-20-(beta-D-glucopyranosyloxy)-dammara-24-ene-3beta-o (2 rxns)

Correction: Stereochemistry added (stereocenters 13->14)

JQOUYGJYNQSCQP-SKWFVUTISA-N -> JQOUYGJYNQSCQP-UZTN0JNJSA-N

BEFORE

AFTER



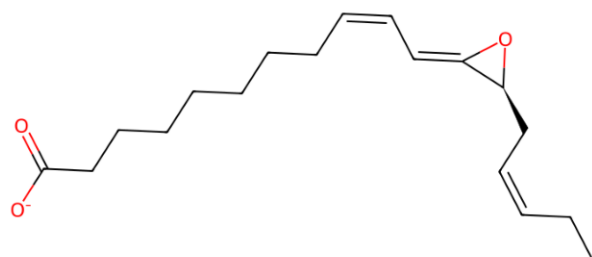
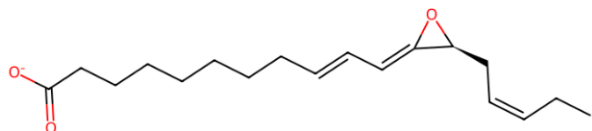
cpd39850 (9Z)-11-[(3S)-3-[(2Z)-pent-2-en-1-yl]oxiran-2-ylidene]undec-9-enoic acid (2 rxns)

Correction: Stereochemistry added (stereocenters 2->3)

YZBZORUZOSCZRN-IRBKMFMFSA-M -> YZBZORUZOSCZRN-YWHLHSFDSA-M

BEFORE

AFTER



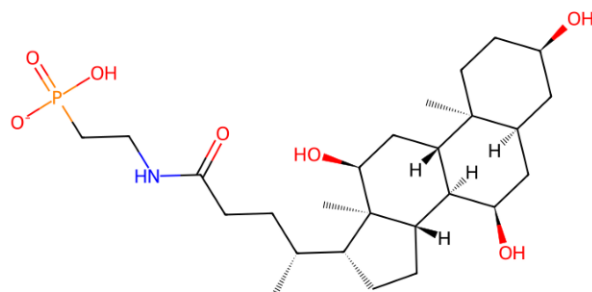
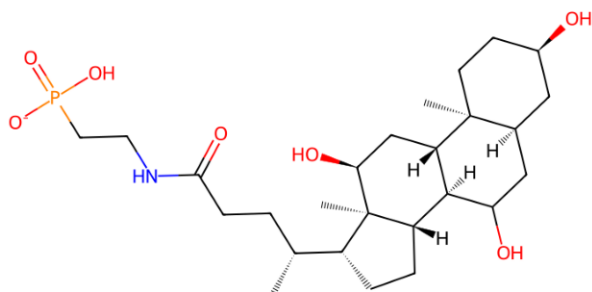
cpd03386 2-Aminoethylphosphocholate (1 rxns)

Correction: Stereochemistry added (stereocenters 10->11)

BGUPNWPECTFDP-RUUYYGZSA-M -> BGUPNWPECTFDP-HZAMXZRMSA-M

BEFORE

AFTER

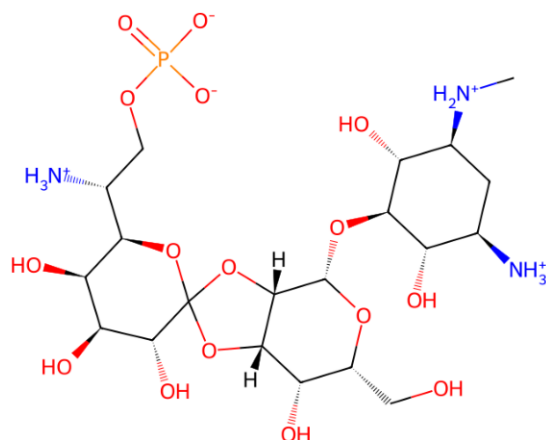


Corrected compounds (visible change) — before / after (28-30 of 58)

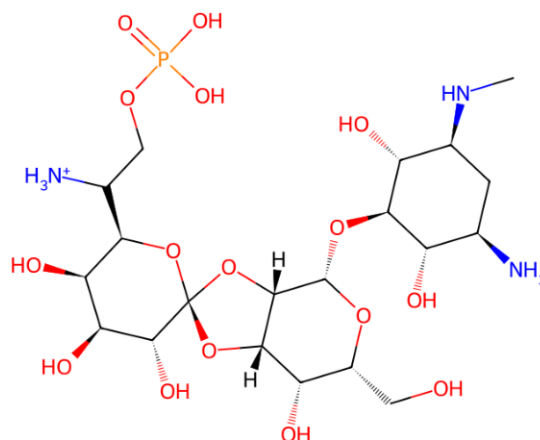
cpd22207 7''-O-phosphohygromycin B (1 rxns)

Correction: Stereochemistry / representation change

DDJWTKQJOKVHBW-GNNWKVJVS-A-0 -> DDJWTKQJOKVHBW-NZSRVPFOSA-0
BEFORE



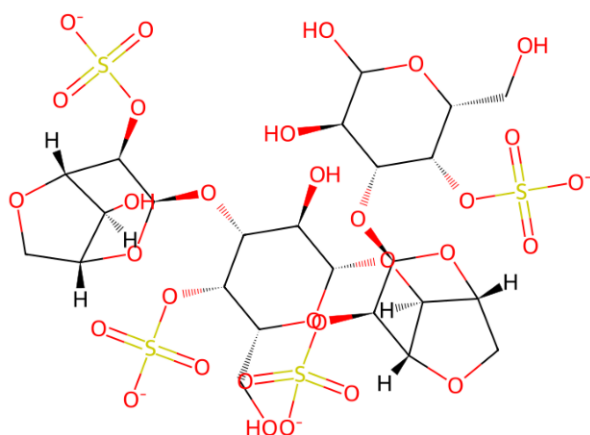
AFTER



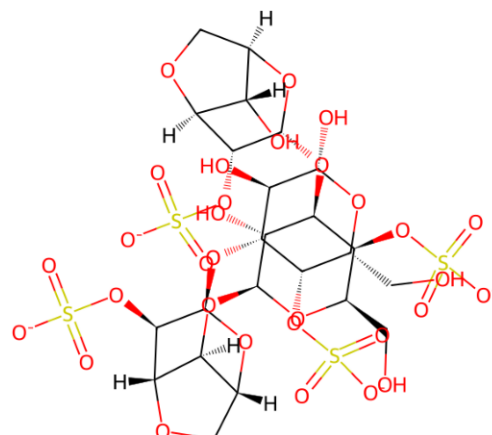
cpd23882 iota-neocarratetraose sulfate (1 rxns)

Correction: Stereochemistry added (stereocenters 19->20)

QILNVJZKGZSQFJ-GCDDTEQSA-J -> QILNVJZKGZSQFJ-IXTYJRGDSA-J
BEFORE



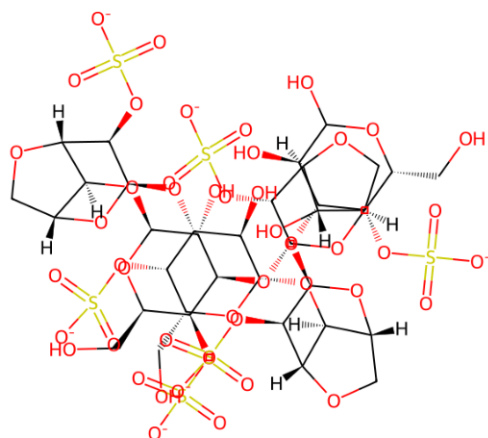
AFTER



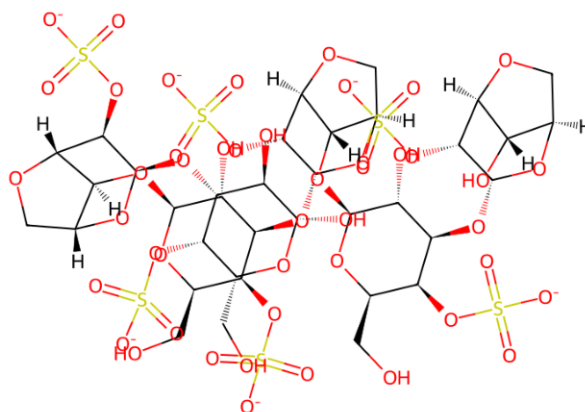
cpd23883 iota-neocarrahexaose sulfate (1 rxns)

Correction: Stereochemistry added (stereocenters 29->30)

JKZMNXQJLUUIRT-ORGYOHQWSA-H -> JKZMNXQJLUUIRT-YTZOPKLZSA-H
BEFORE



AFTER



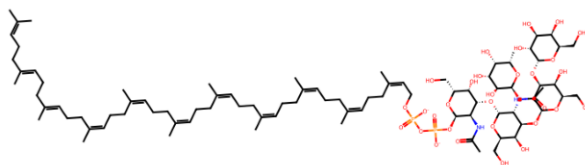
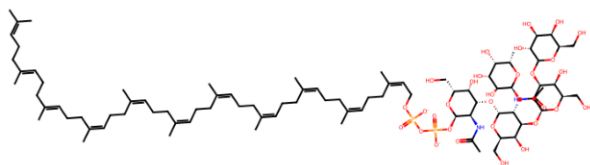
Corrected compounds (visible change) — before / after (31-33 of 58)

cpd31237 α -D-Gal-(1->3)-(α -L-Fuc-(1->2))- β -D-Gal-(1->3)- (1 rxns)

Correction: Stereochemistry added (stereocenters 34->35)

UQYSJTDLRLYPNU-ABPCESQWSA-L -> UQYSJTDLRLYPNU-00XGFABVSA-L
BEFORE

AFTER

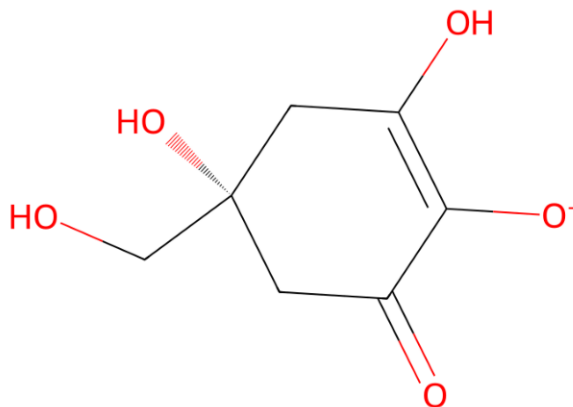
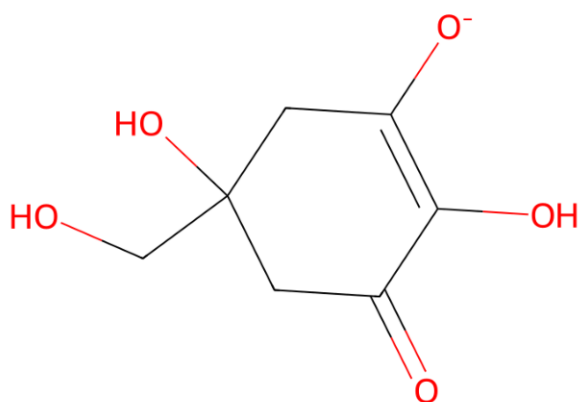


cpd37318 (5S)-2,5-dihydroxy-5-(hydroxymethyl)-3-oxocyclohex-1-en-1- (1 rxns)

Correction: Stereochemistry added (stereocenters 0->1)

0WHGX00DGNBQRG-UHFFFAOYSA-M -> 0WHGX00DGNBQRG-SSDOTTWSA-M
BEFORE

AFTER

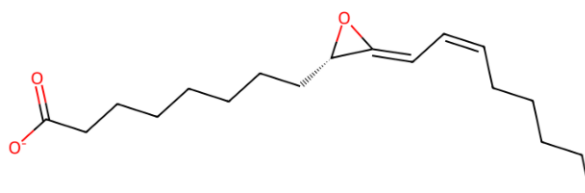
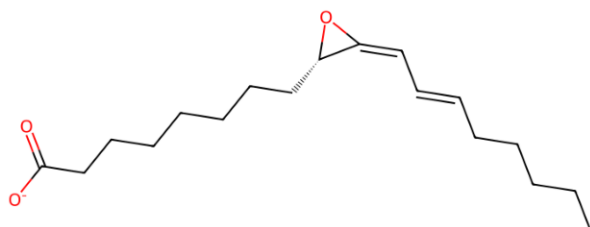


cpd38354 (9S),10-epoxy-(10,12Z)-octadecadienoic acid(1-) (1 rxns)

Correction: Stereochemistry added (stereocenters 1->3)

LVVCD0S0KGYQFY-KRWDZBQ0SA-M -> LVVCD0S0KGYQFY-KKJXIILJSA-M
BEFORE

AFTER



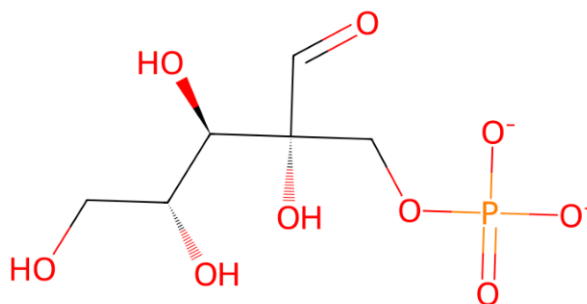
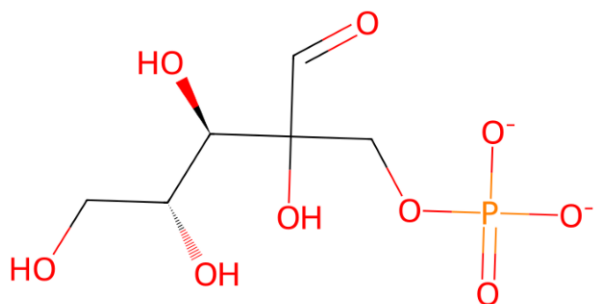
Corrected compounds (visible change) — before / after (34-36 of 58)

cpd39984 2-C-[(phosphonatooxy)methyl]-D-ribose (1 rxns)

Correction: Stereochemistry added (stereocenters 2->3)

RI0ZVCDMYGAYCJ-QYRBDRAASA-L -> RI0ZVCDMYGAYCJ-HSUXUTPPSA-L
BEFORE

AFTER

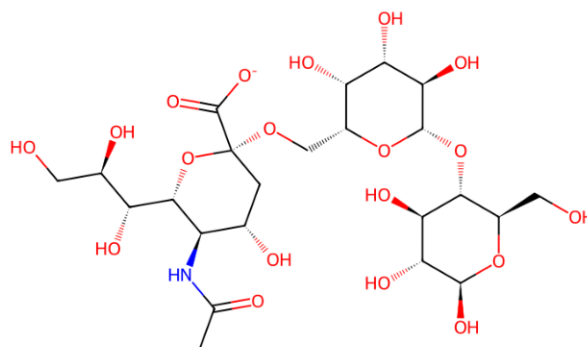
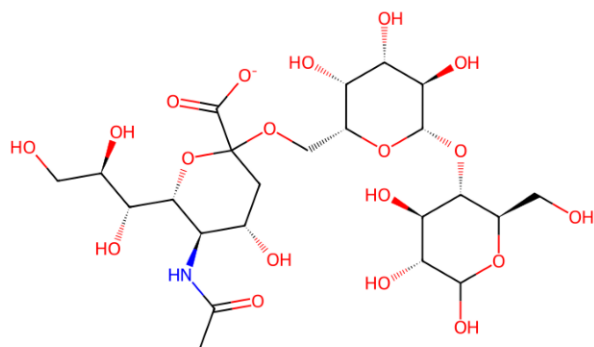


cpd46292 6'-sialyllactose (1 rxns)

Correction: Stereochemistry added (stereocenters 14->16)

TYALNJQZQRNQNQ-SKHIEIRWSA-M -> TYALNJQZQRNQNQ-JLYOMPFMSA-M
BEFORE

AFTER

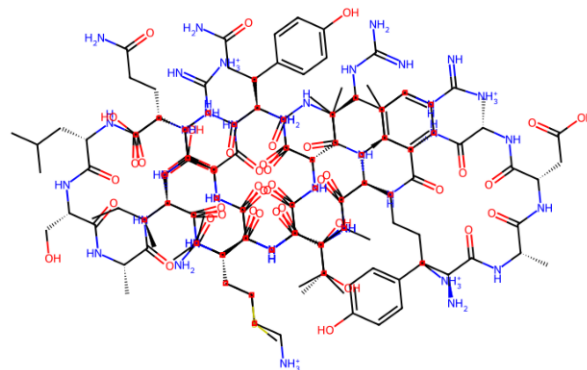
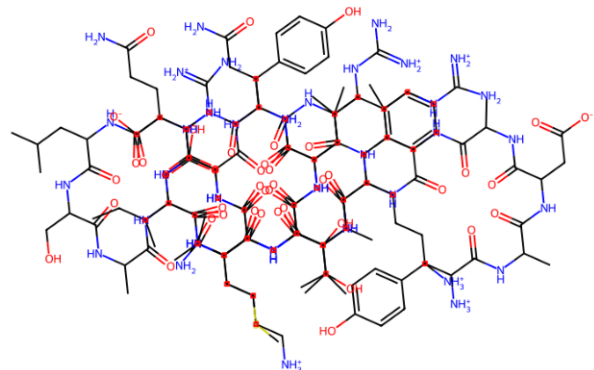


cpd05139 Sermorelin (0 rxns)

Correction: Stereochemistry added (stereocenters 0->31)

WGWRPVFKDLAUQJ-UHFFFAOYSA-R -> WGWRPVFKDLAUQJ-MITYVQBRSA-R
BEFORE

AFTER



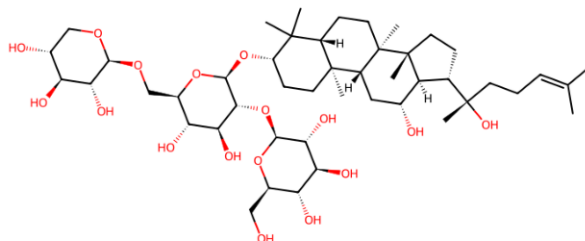
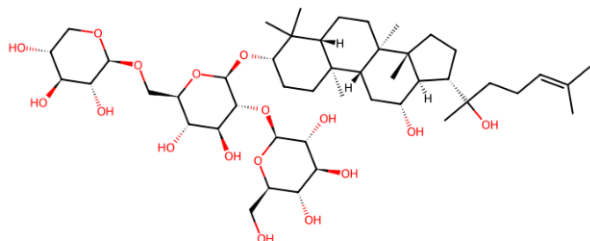
Corrected compounds (visible change) — before / after (37-39 of 58)

cpd17664 Chikusetsusaponin III (0 rxns)

Correction: Stereochemistry added (stereocenters 23->24)

ZICDJJKZDHVLVOD - GREWDSJQSA - N -> ZICDJJKZDHVLVOD - HUGMCNGHSA - N
BEFORE

AFTER

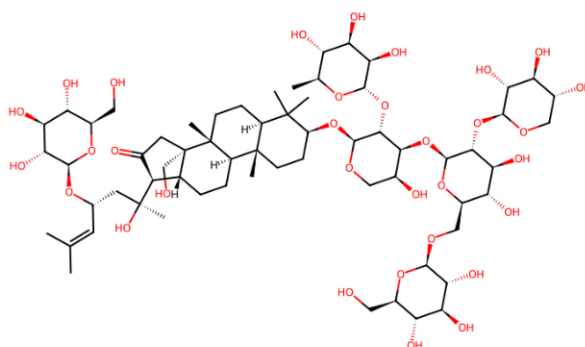
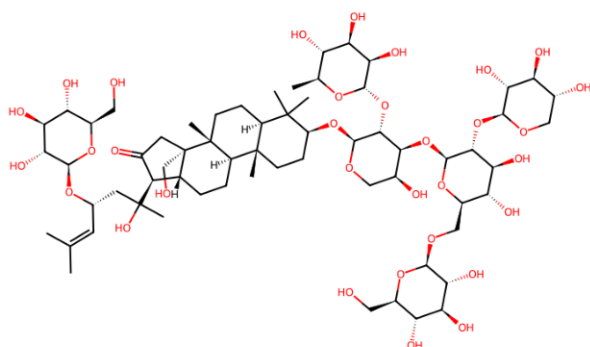


cpd19283 Protojубoside A (0 rxns)

Correction: Stereochemistry added (stereocenters 37->38)

DAYKIQFQZVAKAA - PADDJQGGSA - N -> DAYKIQFQZVAKAA - DJISZYPWSA - N
BEFORE

AFTER

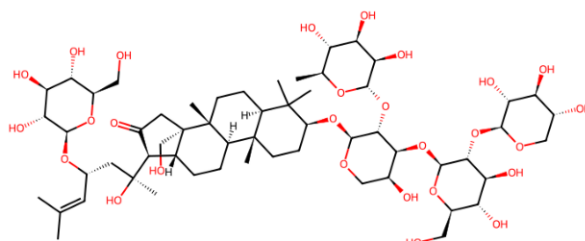
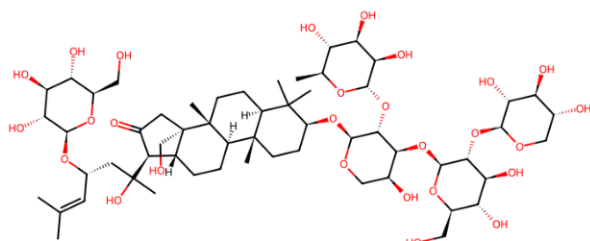


cpd19284 Protojубoside B (0 rxns)

Correction: Stereochemistry added (stereocenters 32->33)

JWDBYPGADNNYPU - RNYOKPNMSA - N -> JWDBYPGADNNYPU - KRACRRDZSA - N
BEFORE

AFTER



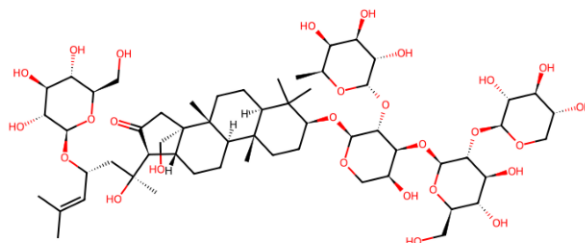
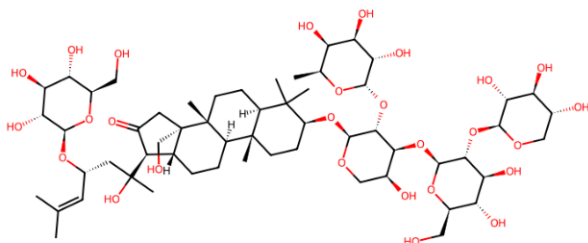
Corrected compounds (visible change) — before / after (40-42 of 58)

cpd19285 Protojubiloside B1 (0 rxns)

Correction: Stereochemistry added (stereocenters 32->33)

JWDBYPGADNNYPU-PNPNNWMTSA-N -> JWDBYPGADNNYPU-HFNHGWQBSA-N
BEFORE

AFTER

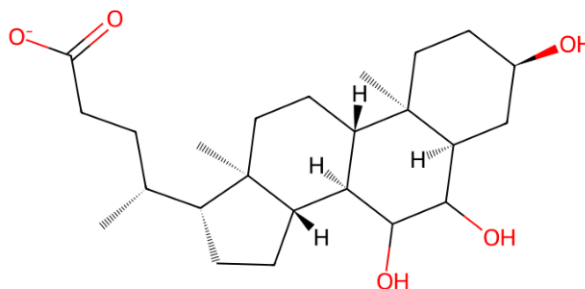
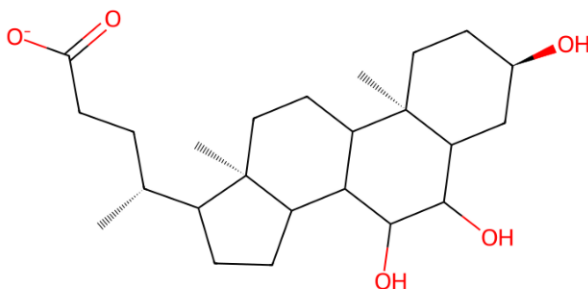


cpd22704 muricholate (0 rxns)

Correction: Stereochemistry added (stereocenters 4->9)

DKPMWHFRUGMUKF-QONOTNIVSA-M -> DKPMWHFRUGMUKF-JDDNAIEOSA-M
BEFORE

AFTER

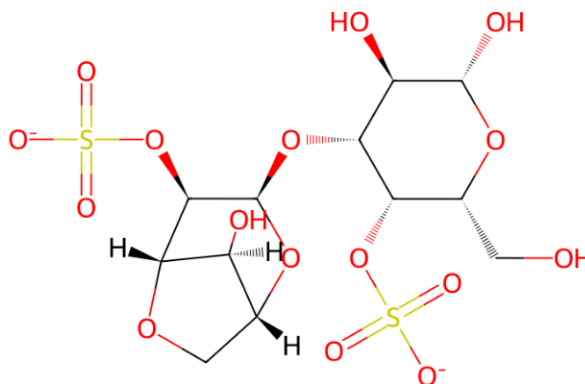
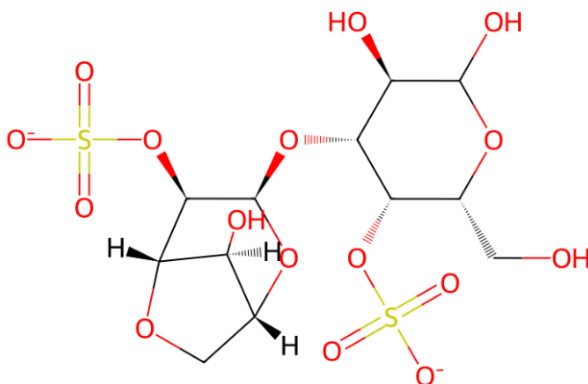


cpd23878 3,6-anhydro-2-O-sulfo-alpha-D-galactopyranosyl-(1->3)-4-O- (0 rxns)

Correction: Stereochemistry added (stereocenters 9->10)

CTNFROXWAZXRKW-VDGMBKLFSA-L -> CTNFROXWAZXRKW-VHBGUFLRSA-L
BEFORE

AFTER

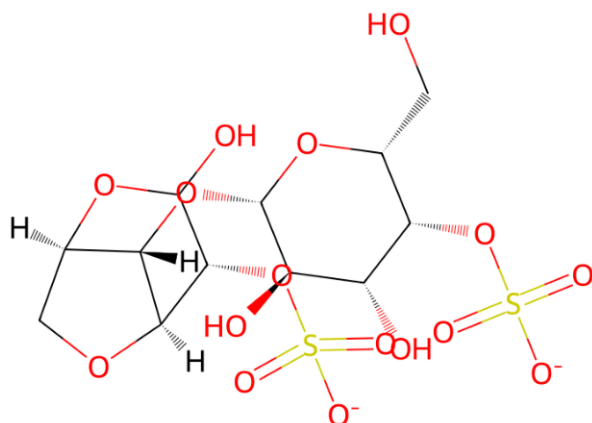


Corrected compounds (visible change) — before / after (43-45 of 58)

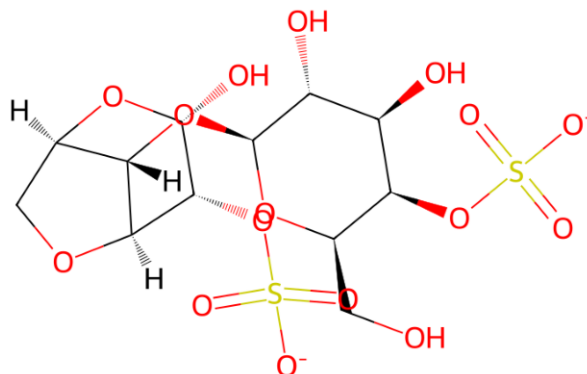
cpd23879 4-O-sulfo- beta-D-galactopyranosyl-(1->4)-3,6-anhydro-2-O- (0 rxns)

Correction: Stereochemistry added (stereocenters 9->10)

MWCCETPFHWJFDS - OSEFXCQPSA - L -> MWCCETPFHWJFDS - XWTDTOCNSA - L
BEFORE



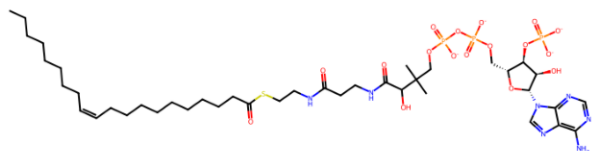
AFTER



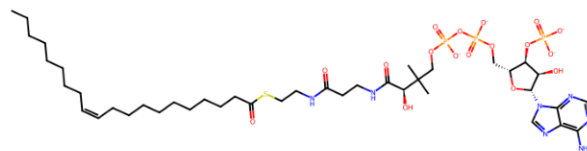
cpd24275 (11Z)-eicos-11-enoyl-CoA (0 rxns)

Correction: Stereochemistry added (stereocenters 5->6)

ZDRKXADSR0CWCG - LURPFOQRSA - J -> ZDRKXADSR0CWCG - FVLDFCIYSA - J
BEFORE



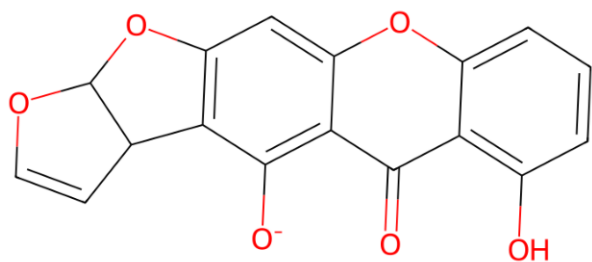
AFTER



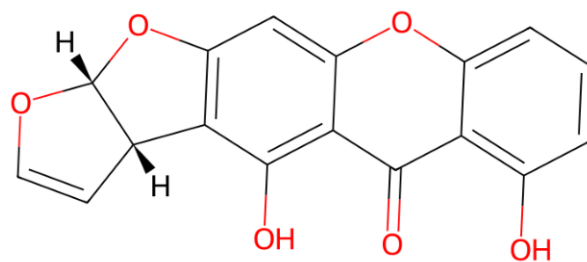
cpd24513 sterigmatin (0 rxns)

Correction: Protonation + stereochemistry (charge -1 -> 0; stereocenters 0->2)

CWYJYLXZMAUSNI - UHFFFAOYSA - M -> CWYJYLXZMAUSNI - BWKAKNAASA - N
BEFORE



AFTER

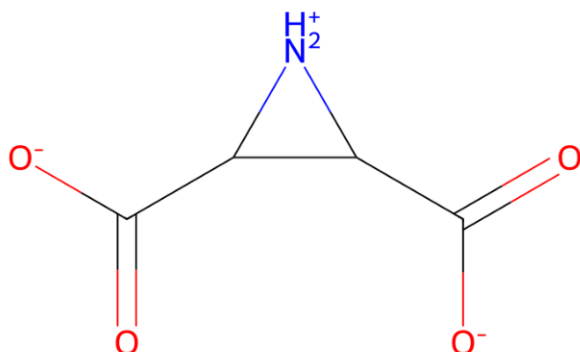


Corrected compounds (visible change) — before / after (46-48 of 58)

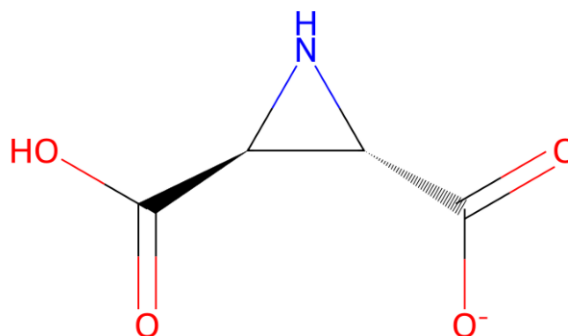
cpd26249 S-2,3-dicarboxyaziridine (0 rxns)

Correction: Stereochemistry added (stereocenters 0->2)

IFCCPDAHQDGHMH-UHFFFAOYSA-M -> IFCCPDAHQDGHMH-LWMBPPNESA-M
BEFORE



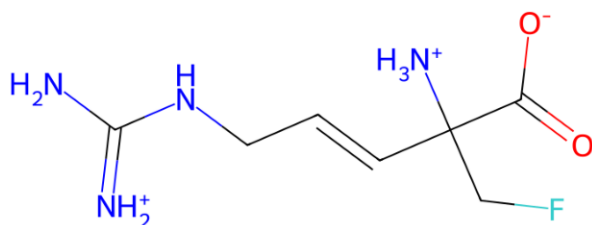
AFTER



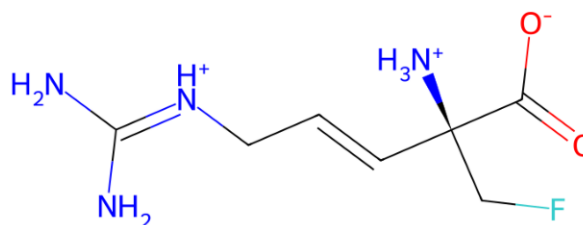
cpd26275 (E)-alpha-monofluoromethyl-3,4-dehydroarginine (0 rxns)

Correction: Stereochemistry added (stereocenters 1->2)

UDWPEHOXJIIIEY-OWOJBTEDSA-0 -> UDWPEHOXJIIIEY-Q0HHWTFISA-0
BEFORE



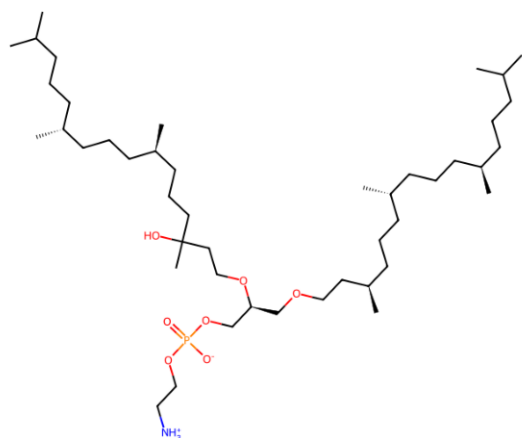
AFTER



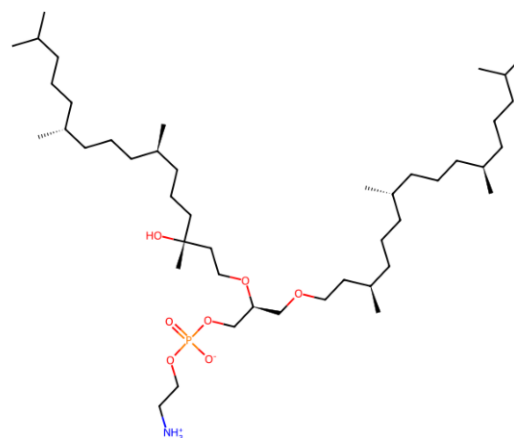
cpd31660 2-O-(3'-hydroxy)phytanyl-3-O-phytanyl-sn-glycero-1-phospho (0 rxns)

Correction: Stereochemistry added (stereocenters 6->7)

VUKYXYWCFFKPMX-PLSZJGEGSA-N -> VUKYXYWCFFKPMX-ARNVMZJBSA-N
BEFORE



AFTER



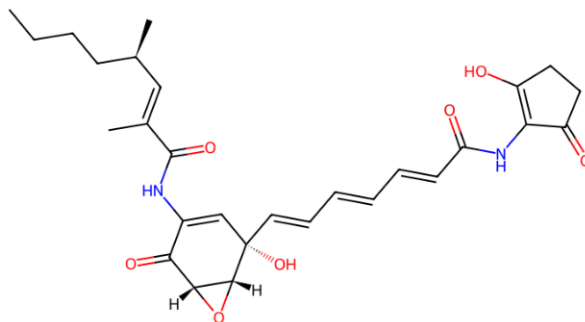
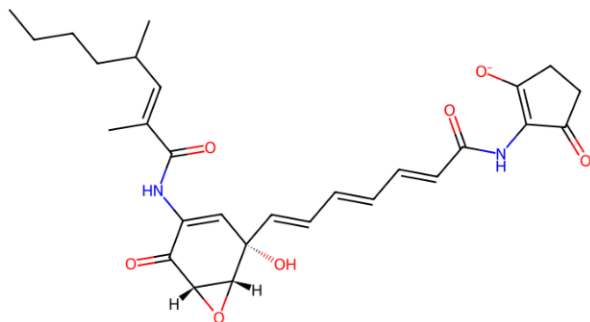
Corrected compounds (visible change) — before / after (49-51 of 58)

cpd34026 manumycin B (0 rxns)

Correction: Protonation + stereochemistry (charge -1 -> 0; stereocenters 7->8)

ZGICGDCGECBVTD-ULN0GCHZSA-M -> ZGICGDCGECBVTD-L0WJZYKPSA-N
BEFORE

AFTER

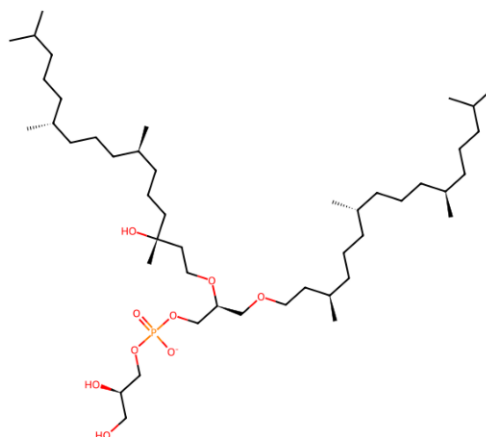
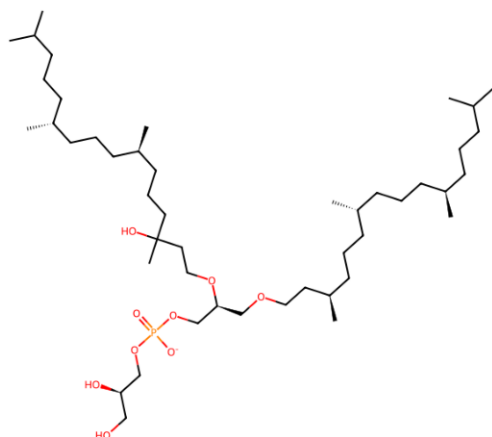


cpd34426 2-O-(3'-hydroxy)phytanyl-3-O-phytanyl-sn-glycerol-1-phosph (0 rxns)

Correction: Stereochemistry added (stereocenters 7->8)

BIVFMXPCWM0NSY-XMRRTHDVSA-M -> BIVFMXPCWM0NSY-XHQCNPPWSA-M
BEFORE

AFTER

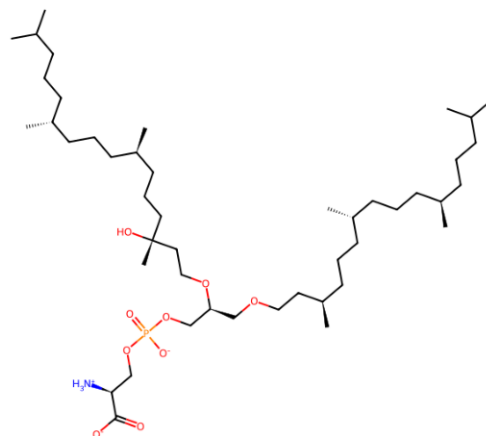
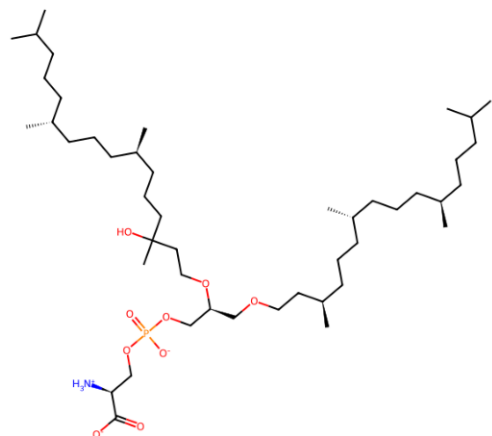


cpd36034 2-O-(3'-hydroxy)phytanyl-3-O-phytanyl-sn-glycero-1-phospho (0 rxns)

Correction: Stereochemistry added (stereocenters 7->8)

YALRQ0MGANIIQK-B0MFZTBMSA-M -> YALRQ0MGANIIQK-QQLPJPLTSA-M
BEFORE

AFTER



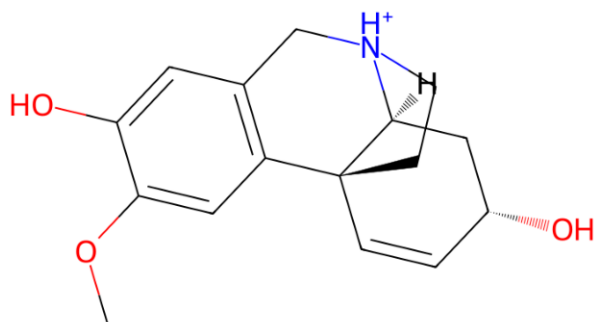
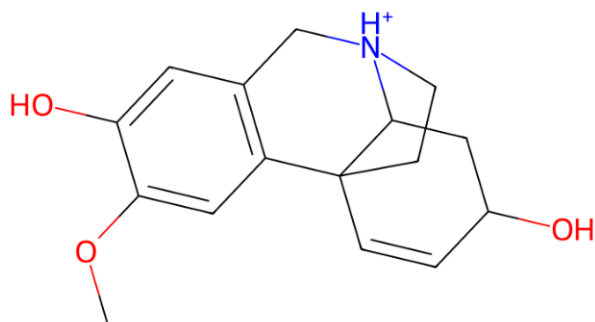
Corrected compounds (visible change) — before / after (52-54 of 58)

cpd44629 normaritidine (0 rxns)

Correction: Stereochemistry added (stereocenters 0->3)

TZTBAJFJEZRQCV-UHFFFAOYSA-0 -> TZTBAJFJEZRQCV-RLCCDNCMSA-0
BEFORE

AFTER

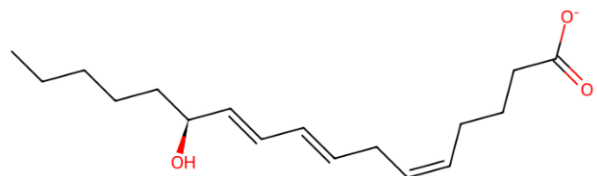
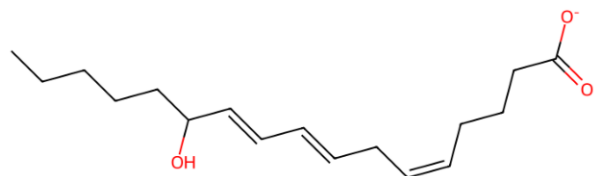


cpd44677 (5Z,8E,10E)-12-hydroxyheptadeca-5,8,10-trienoate (0 rxns)

Correction: Stereochemistry added (stereocenters 3->4)

KUKJHGXXZWHSBG-FQWDWNLFSA-M -> KUKJHGXXZWHSBG-WBGSEQOASA-M
BEFORE

AFTER

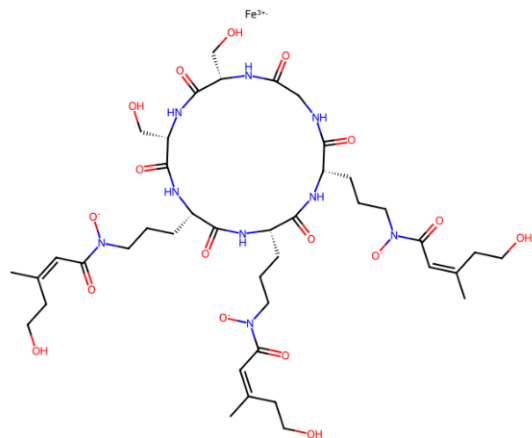
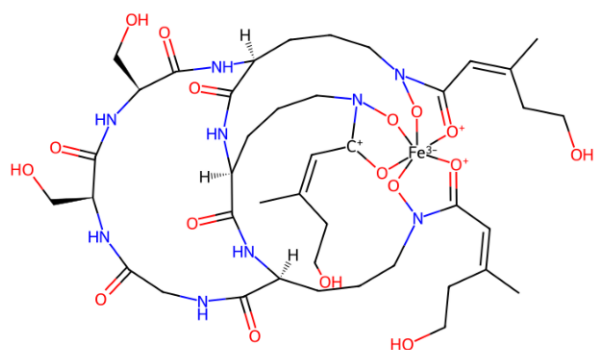


cpd46464 ferrirhodin (0 rxns)

Correction: Stereochemistry added (stereocenters 5->8)

SEQVPEQKJMMKN0-SQLDAFDYSA-N -> SEQVPEQKJMMKN0-RQCMKQRDSA-N
BEFORE

AFTER

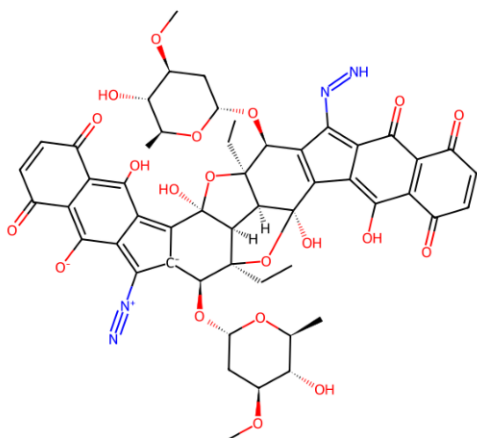


Corrected compounds (visible change) — before / after (55-57 of 58)

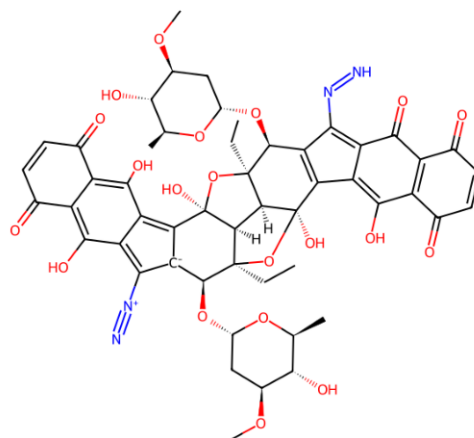
cpd46646 lomaiviticin B (0 rxns)

Correction: Protonation corrected (charge -1 -> 0)

ZPEQGULUANUJNH-RJEFGBVDSA-N -> ZPEQGULUANUJNH-RJEFGBVDSA-O
BEFORE



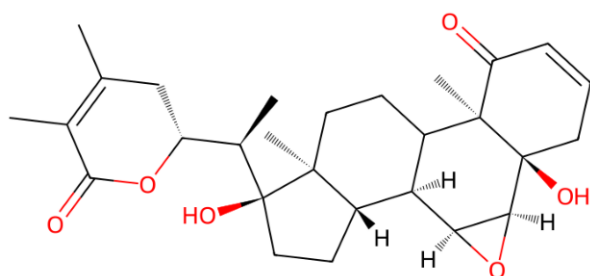
AFTER



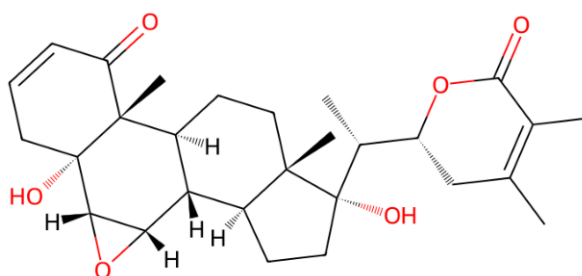
cpd46783 withanone (0 rxns)

Correction: Stereochemistry added (stereocenters 10->11)

FAZIYUIDUNHZRG-QUSHIGSFSA-N -> FAZIYUIDUNHZRG-PCTWTJKKSA-N
BEFORE



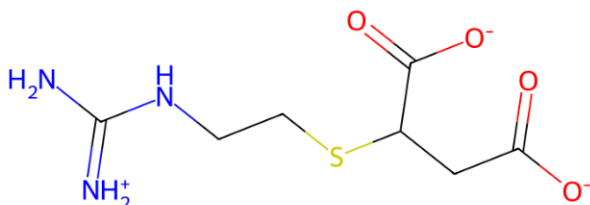
AFTER



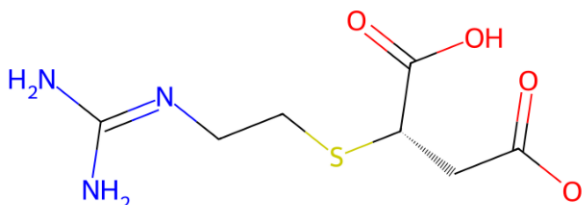
cpd47191 [(2-guanidinoethyl)sulfanyl]butanedioate (0 rxns)

Correction: Stereochemistry added (stereocenters 0->1)

VKVCLXDFQQAABP-UHFFFAOYSA-M -> VKVCLXDFQQAABP-BYPYZUCNSA-M
BEFORE



AFTER



Corrected compounds (visible change) — before / after (58-58 of 58)

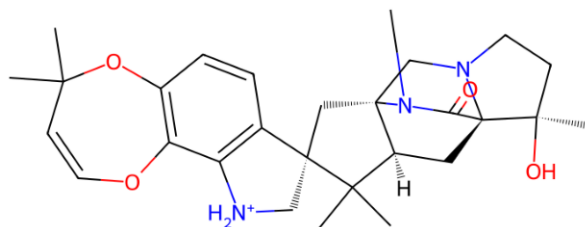
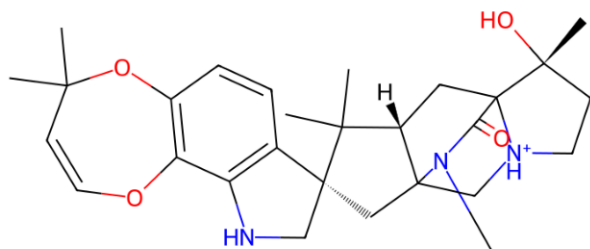
cpd47672 2-deoxyparaherquamide A (0 rxns)

Correction: Stereochemistry added (stereocenters 3->5)

DYVLXWPZFQQIUU-BUWOMYMCSA-0 -> DYVLXWPZFQQIUU-WGNDVSEMSA-0

BEFORE

AFTER



Newly added compounds (40)

compounds with no prior entry in Unique_ModelSEED_Structures.txt

Newly added compounds (1-4 of 40)

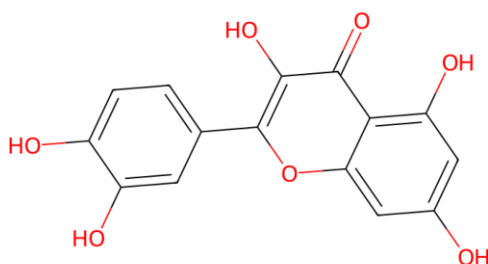
cpd00074 16S (65 rxns)

InChIKey RWSOTUBLDIXVET-UHFFFA0YSA-N



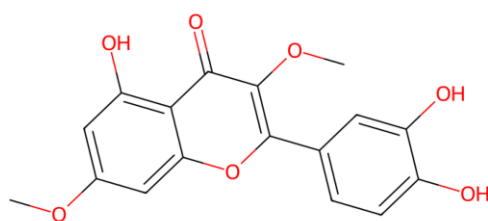
cpd00313 2-(3,4-dihydroxyphenyl)-3,5-dihydroxy-4-oxo-4H-chromen-7 (46 rxns)

InChIKey REFJWTPEDVJJIY-UHFFFA0YSA-N



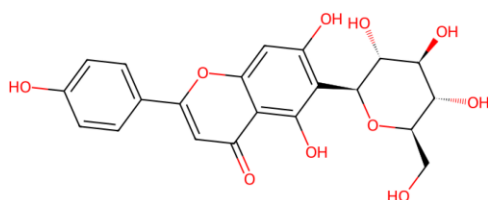
cpd00928 2-(3,4-dihydroxyphenyl)-3,7-dimethoxy-4-oxo-4H-chromen- (4 rxns)

InChIKey LUJAXSNNYBCFEE-UHFFFA0YSA-N



cpd01180 (1S)-1,5-anhydro-1-[5-hydroxy-2-(4-hydroxyphenyl)-7-oxi (13 rxns)

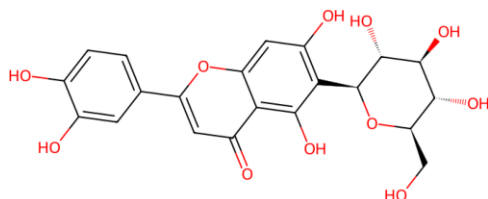
InChIKey MYXNKGACZJSMBT-VJXVFPJBSA-N



Newly added compounds (5-8 of 40)

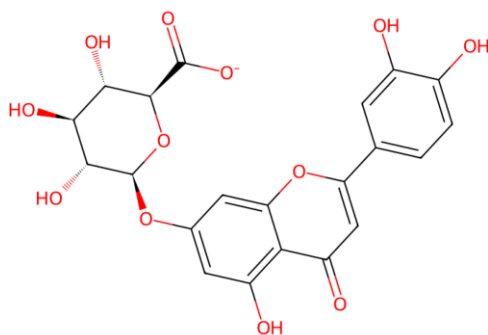
cpd01254 (1S)-1,5-anhydro-1-[2-(3,4-dihydroxyphenyl)-5-hydroxy-7 (2 rxns)

InChIKey ODBRNZZJSYPIDI-VJXVFPJBSA-N



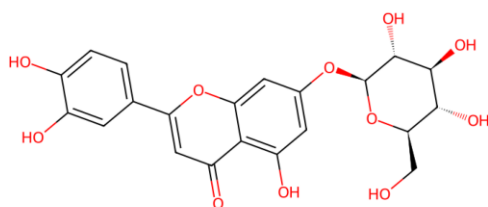
cpd02215 2-(3,4-dihydroxyphenyl)-7-(beta-D-glucopyranuronosyloxy (4 rxns)

InChIKey VSUOKLTVXQRUSG-ZFORQUDYSA-M



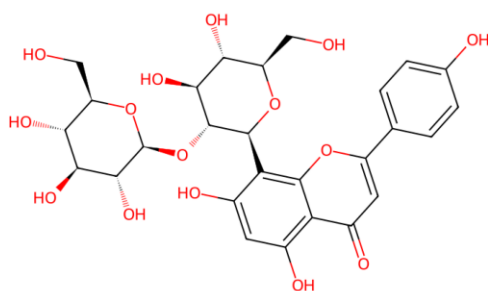
cpd02453 2-(3,4-dihydroxyphenyl)-7-(beta-D-glucopyranosyloxy)-4- (6 rxns)

InChIKey PEFNSGRTCBGNAN-QNDFHXLGSA-N



cpd02492 (1S)-1,5-anhydro-2-O-beta-D-glucopyranosyl-1-[5-hydroxy (2 rxns)

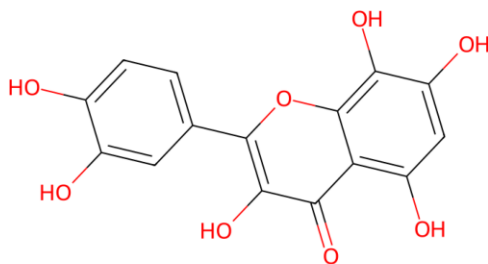
InChIKey FYT0THFWELW0CG-0AFKVUNRSA-N



Newly added compounds (9-12 of 40)

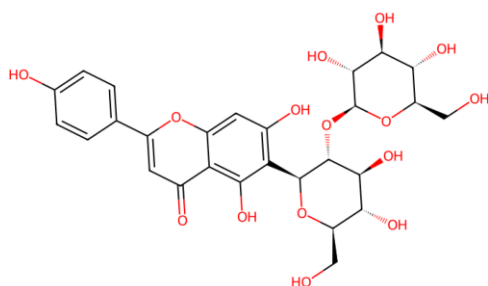
cpd02539 2-(3,4-dihydroxyphenyl)-3,5,7,8-tetrahydroxy-chromen-4- (3 rxns)

InChIKey YRRAGUMVDQZIIY-UHFFFAOYSA-N



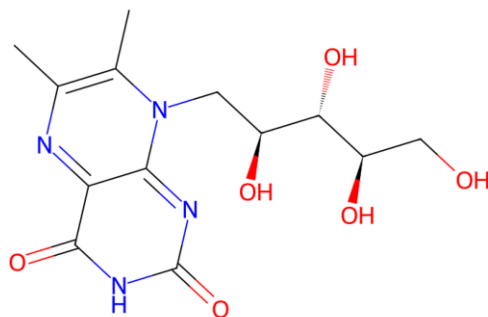
cpd02579 Isovitexin 2''-O-beta-D-glucoside (3 rxns)

InChIKey RQTTXGQDIROLTQ-FASGCTRLSA-N



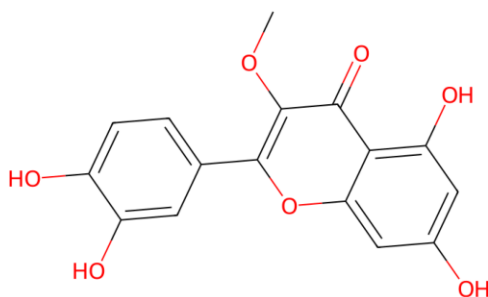
cpd02656 1-deoxy-1-(6,7-dimethyl-2,4-dioxo-2H-pteridin-3-yl)-D-glucitol (12 rxns)

InChIKey SXDXRJZUAJBNFL-XKSSXDPKSA-N



cpd02712 2-(3,4-dihydroxyphenyl)-5-hydroxy-3-methoxy-4-oxo-4H-ch (3 rxns)

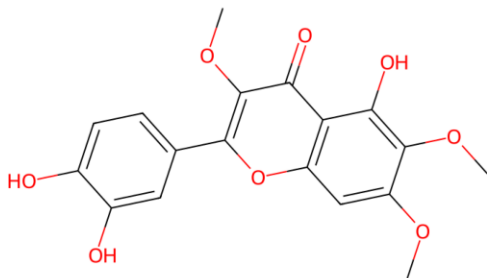
InChIKey WEPBGSIWZTEJR-UHFFFAOYSA-N



Newly added compounds (13-16 of 40)

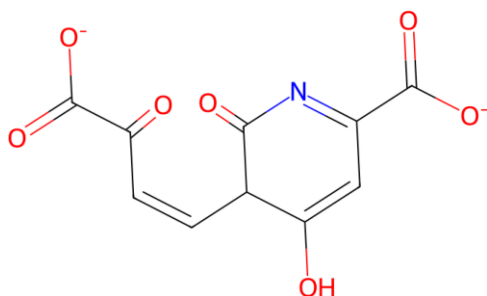
cpd02771 2-(3,4-dihydroxyphenyl)-5-hydroxy-3,6,7-trimethoxy-4H-c (2 rxns)

InChIKey BYWLLSQTJBXAPV-UHFFFAOYSA-N



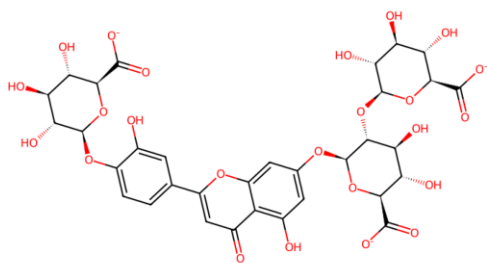
cpd02931 5-(3'-Carboxy-3'-oxopropenyl)-4,6-dihydroxypicolinate (3 rxns)

InChIKey OBMUVUBVUPSHMJ-UPHRSURJSA-L



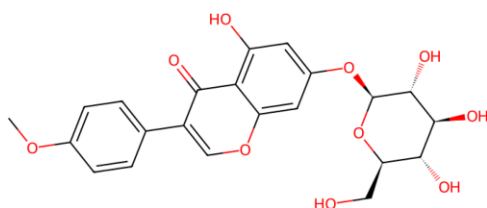
cpd02981 7-{[2-O-(beta-D-glucopyranuronosyl)-beta-D-glucopyranur (3 rxns)

InChIKey AEYXZGCDWDUIKX-0FFAAIFBSA-K



cpd03184 7-(beta-D-glucopyranosyloxy)-3-(4-methoxyphenyl)-4-oxo- (7 rxns)

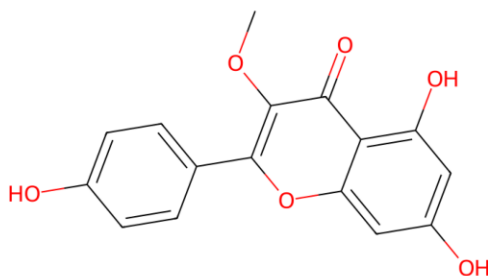
InChIKey LFEUICHQZGNOHD-RECXWPGBSA-N



Newly added compounds (17-20 of 40)

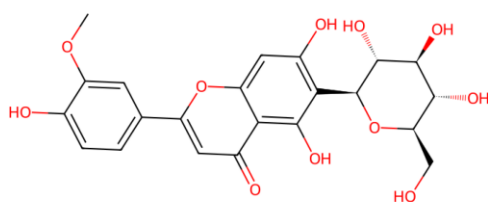
cpd03499 3-O-methylkaempferol (2 rxns)

InChIKey VJJZJBUCDWKPLC-UHFFFA0YSA-N



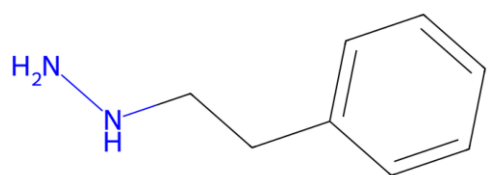
cpd03565 (1S)-1,5-anhydro-1-[5-hydroxy-2-(4-hydroxy-3-methoxyphenoxy)-6-hydroxy-3-methoxyphenoxy]hexane (2 rxns)

InChIKey KOMUHHCFAXYRPO-DGHBABESA-N



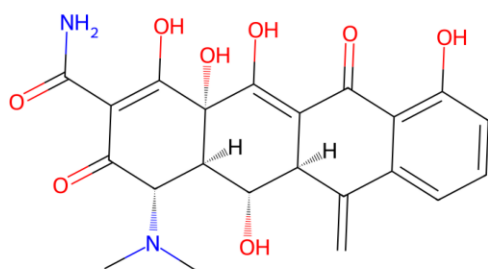
cpd04612 Fenelzyna (0 rxns)

InChIKey RMUCZJUITONUFY-UHFFFA0YSA-N



cpd04815 6-demethyl-6-deoxy-5-hydroxy-6-methylenetetraacycline (0 rxns)

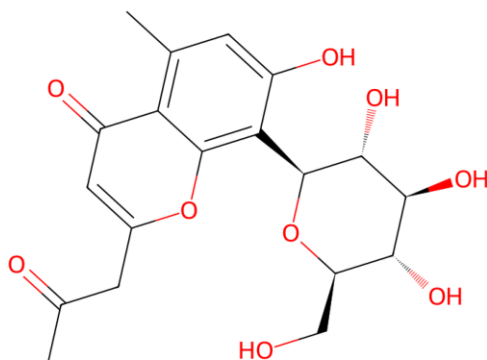
InChIKey ZZN RJDRMIATXIN-IWV LMIASSA-N



Newly added compounds (21-24 of 40)

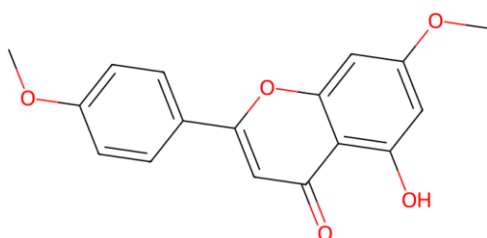
cpd05890 Aloesin (2 rxns)

InChIKey HKIKAXXIWJHWLY-ZIIYPAMZSA-N



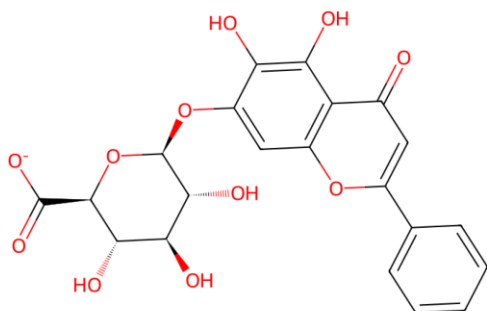
cpd06909 4',7-dimethylapigenin (2 rxns)

InChIKey LZERJKGWTQYMBB-UHFFFAOYSA-N



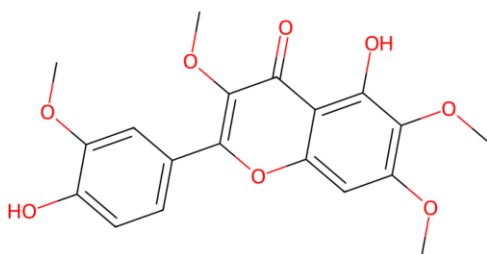
cpd06915 5,6,7-trihydroxyflavone 7-O-beta-D-glucuronate (4 rxns)

InChIKey IKIIZLYTISPENI-ZFORQUDYSA-M



cpd06920 3,6,7,3'-tetra-methylquercetagenin (1 rxns)

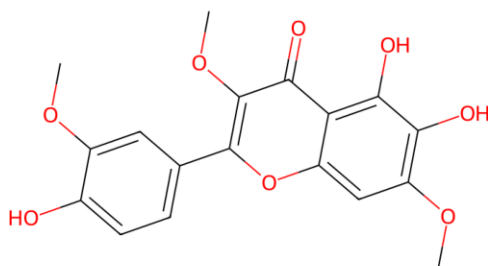
InChIKey NBVTYGIYKCPHQN-UHFFFAOYSA-N



Newly added compounds (25-28 of 40)

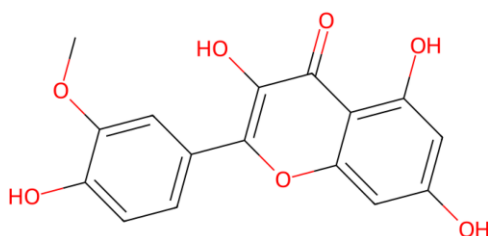
cpd06921 3, 7, 3'-O-trimethylquercetagenin (2 rxns)

InChIKey QQBSPLCHDUCBNM-UHFFFA0YSA-N



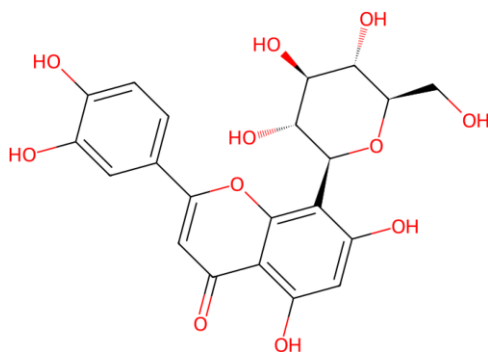
cpd06974 3'-methoxyquercetin (6 rxns)

InChIKey IZQSVPB0UDKVDZ-UHFFFA0YSA-N



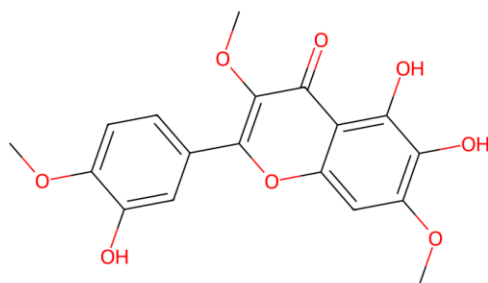
cpd07003 2-(3,4-dihydroxyphenyl)-5,7-dihydroxy-6-[(2S,3R,4R,5S,6 (7 rxns)

InChIKey PLAPMLGJVGLZ0V-VPRICQMDSA-N



cpd07005 3,7,4'-trimethylquercetagenin (4 rxns)

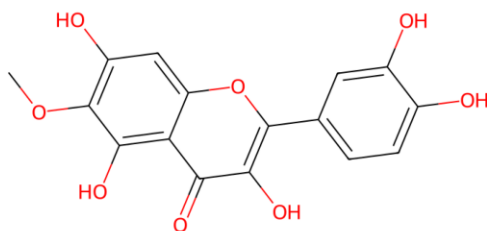
InChIKey UQBUUCDVKDSHCE-UHFFFA0YSA-N



Newly added compounds (29-32 of 40)

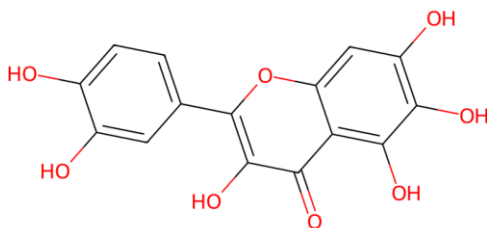
cpd07007 6-methoxyquercetin (4 rxns)

InChIKey JMIFIYIEXODVTO-UHFFFAOYSA-N



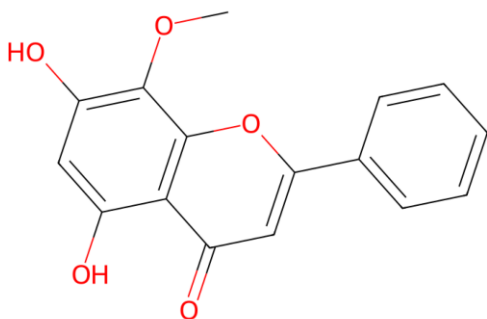
cpd07011 6-Hydroxyquercetin (3 rxns)

InChIKey ZV0LCUVKHLEPEV-UHFFFAOYSA-N



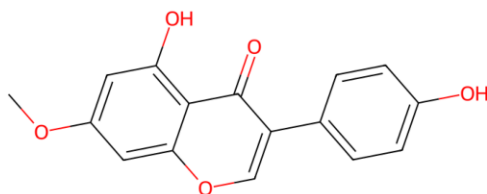
cpd07086 5,7-dihydroxy-8-methoxy-2-phenyl-4H-1-benzopyran-4-one (3 rxns)

InChIKey XLTFNNCXVBYBSX-UHFFFAOYSA-N



cpd07407 3-(4-hydroxyphenyl)-7-methoxy-4-oxo-4H-1-benzopyran-5-o (4 rxns)

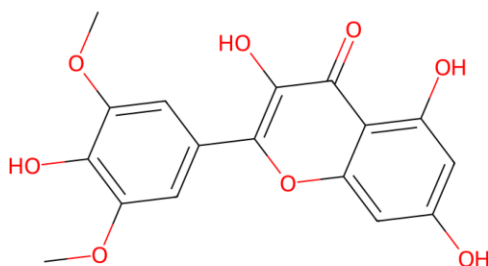
InChIKey KQMVAGISDHMXJJ-UHFFFAOYSA-N



Newly added compounds (33-36 of 40)

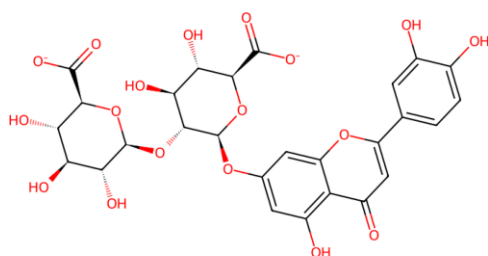
cpd08434 3',5'-O-Dimethylmyricetin (5 rxns)

InChIKey UZMAPBJVXOG0FT-UHFFFAOYSA-N



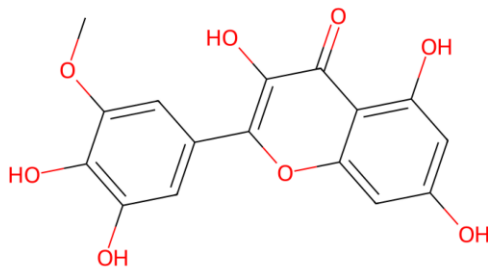
cpd09263 2-(3,4-dihydroxyphenyl)-7-[[2-O-(beta-D-glucopyranurono (6 rxns)

InChIKey PBBVWJQPAZYQDB-DBFWEQBMSA-L



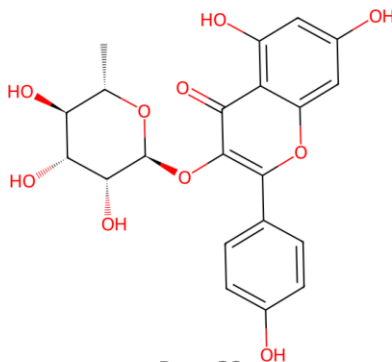
cpd09264 2-(3,4-dihydroxy-5-methoxyphenyl)-5,7-dihydroxy-4-oxo-4 (4 rxns)

InChIKey CFYMYCCYMJIYAB-UHFFFAOYSA-N



cpd17189 3-[(6-deoxy-alpha-L-mannopyranosyl)oxy]-5-hydroxy-2-(4- (1 rxns)

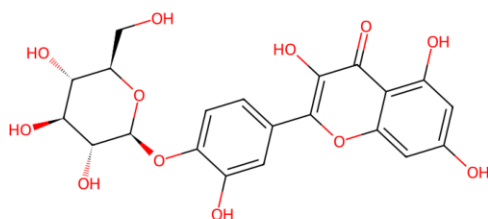
InChIKey SOSLMHZ0JATCCP-AEIZVZFYSA-N



Newly added compounds (37-40 of 40)

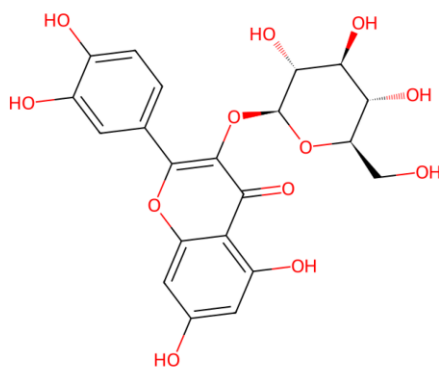
cpd23057 quercetin 4'-O-beta-D-glucoside (6 rxns)

InChIKey 0IUBYZLTFSLSBY-HMGRVEA0SA-N



cpd26557 2-(3,4-dihydroxyphenyl)-3-(beta-D-glucopyranosyloxy)-5- (25 rxns)

InChIKey 0VSQVDMCBVZWGM-QS0FNFLRSA-N



cpd27013 16S (0 rxns)

InChIKey RWSOTUBLDIXVET-UHFFFA0YSA-N



cpd27311 16S (0 rxns)

InChIKey RWSOTUBLDIXVET-UHFFFA0YSA-N

